FINAL REPORT · PUERTO VALLARTA · MONDAY 12 OCTOBER 2026, 8 AM

Hurricane Simon and Puerto Vallarta: what was forecast, what happened, and what coastal early warning needs next

Every forecast made for the city during the storm, set against the outcome; a verification of 25 physics, AI and ensemble models; a reply to the uploaded analysis *Comprehensive Verification and Theoretical Analysis of Dynamical and AI-Driven Tropical Cyclone Forecasting*; and a proposal for how a public-private partnership could give this coast the forecasting it will need as the climate and the models change.

Based on NHC advisories 1–20A (last: 6 am CST Monday) · SMN aviso 40 (6 am CST) · ATCF decks to Monday 6 am · ECMWF and NOAA open data · NOAA/CIRA AI-model archive · METAR MMPR to Monday 8 am · GOES-18 · residents' reports

> **Not an official forecast.** Follow the SMN, the NHC and Protección Civil. Emergencies: 911. Everything here is an independent reading of public data; it is meant to help you interpret the official products, not replace them.

1. The short version

- **What happened in Vallarta: nothing.** Simon's center passed its closest point to the city, 120 km to the south-west, around midnight Sunday as a 95-kt hurricane, then stalled. The airport reported winds of 0–4 kt all night and light rain on and off; residents in the city reported calm and little rain. The eastern eyewall brushed the Jalisco coast between Playa Pérula and Cabo Corrientes, where the hurricane warning remains.
- **What was forecast.** From Thursday morning to Saturday night (advisories 5 to 15) the NHC line put the center within 20 km of the city on ten of those eleven advisories (41 km on advisory 6), as a hurricane of category 1 to 3 at that point on most of them. The line moved to 50 km on Sunday 3 am, 74 km at 9 am and 101 km at 3 pm; the real closest pass was 120 km. The NHC's own odds of tropical-storm-force wind in the city were 56–76% from Friday to Sunday morning and dropped to 8% at 3 pm Sunday.
- **What our resident graphics said.** The sets built on Sunday morning gave the three buildings gusts of 68–83 km/h; the 3 pm set gave 47–61 km/h; the first set also gave 200–250 mm of rain, scaled to the SMN regional figure, before it was corrected the same day on a ground report that it was not raining. None of this happened. The morning numbers came from the NHC's own 56% probability of tropical-storm-force wind, converted into the wind with a 50% chance and scaled to each building; the 3 pm numbers came from the NHC line and its open-water wind envelope. Neither came from the models that were right.

- **Which models were right.** Over the two days before the pass, the best position forecasts came from Google DeepMind's cyclone model (mean error 33 km), the NHC's corrected consensus (38 km) and the NHC official forecast (47 km). The US regional hurricane models and UKMET carried the center about 150 km too far north; the GFS-driven AI models did as badly on the leads they could be checked. Every gridded AI model collapsed Simon's intensity to a tropical storm while it stayed a category 4.
- **What comes next.** Advisory 20A (Monday 6 am): the center is 80 km west-southwest of Cabo Corrientes, drifting northwest at 4 km/h; it is forecast to move slowly away from the bay, then turn northeast toward the Nayarit–Sinaloa coast around Wednesday night as a weakening hurricane. The NHC's Vallarta probabilities are now below the 3% display threshold. The SMN (aviso 40) forecasts 50–75 mm for western Jalisco and the Nayarit coast and waves of 4–5 m on the Jalisco coast, decreasing through the day. The sea remains the hazard that is certain.
- **On the uploaded analysis.** Its physics and its account of the AI intensity problem hold up and Simon confirmed them. Its model-verification table does not: the numbers have no source, and several contradict the record. It also omits the one model that actually won.
- **What this storm asks for.** The gap was not in the global models; it was in everything between the global grid and a window in Amapas: no radar over Banderas Bay, no public verification of the forecast as it ran, no local wind or rain model trained on local observations, and a cone that cannot carry probability. Section 10 sets out a public-private partnership that could close that gap in three seasons, with the roles, the money and the measures of success.

2. Monday morning: where Simon is and what is forecast

NHC advisory 20A (6 am CST Monday): center 20.1N 106.4W, 100 kt (115 mph), 958 mb, drifting northwest at 4 km/h, 80 km west-southwest of Cabo Corrientes. The forecast (advisory 20): 20.6N 106.7W and 90 kt by noon Monday, 21.2N 107.1W and 85 kt by midnight, 21.8N 107.3W and 80 kt by noon Tuesday, 80 kt through Wednesday noon near 22.9N 107.0W, then inland near 23.4N 106.6W as a 70-kt hurricane around midnight Wednesday and dissipating over the Sierra Madre by Friday. The discussion notes a higher-than-normal track uncertainty while the storm sits between two ridges, and that the latest line was moved slightly west "after considering some of the latest AI-based guidance". Puerto Vallarta no longer appears in the wind-speed probability table, which lists a location only when its chance of tropical-storm-force wind reaches 3%.

SMN aviso 40 (6 am CST): very heavy point rainfall (50–75 mm) in western Jalisco, the Nayarit coast and west, and southern Sinaloa; gusts of 90–100 km/h and waves of 4–5 m on the Jalisco coast, decreasing through the day; gusts of 50–60 km/h and waves of 2.5–3.5 m on the coasts of Nayarit and central and southern Sinaloa. The NHC's rainfall statement keeps 2–4 in (50–100 mm) of additional rain across coastal southwest and west-central Mexico, isolated 6–8 in in Jalisco, and isolated storm totals of 30 in over parts of Michoacán, Colima and Jalisco.

For the city today: the wind the parametric field gives the three buildings is now 37–47 km/h in gusts from the south-east, the sea is the remaining hazard, and the rain picture changes if the storm lingers to the west long enough for the wind to turn onshore (the consensus puts that on Tuesday morning). The building summary for advisory 20 is Figure 1; it replaces the Sunday 3 pm summary, which is kept in Appendix A for the record.

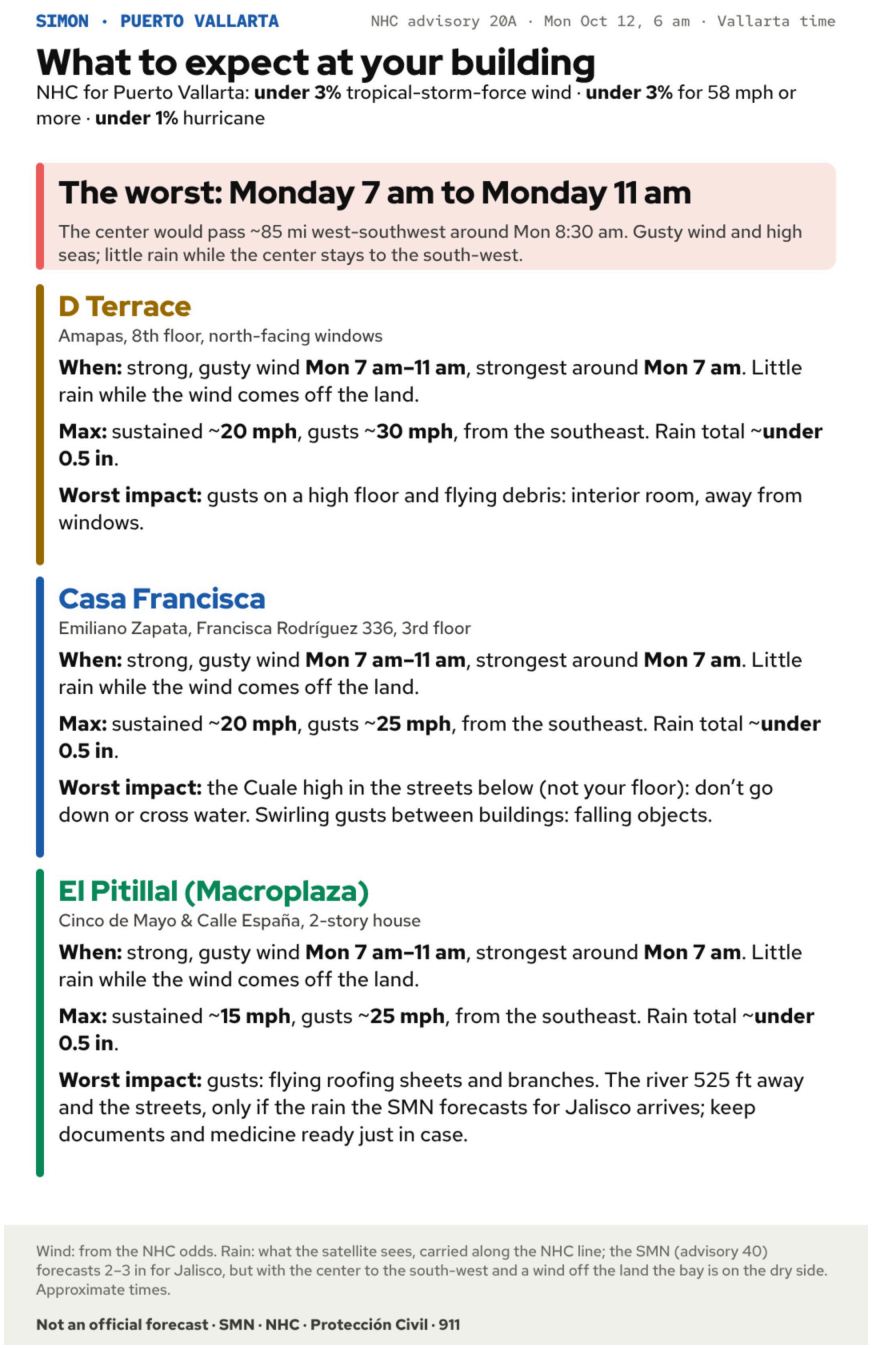


Figure 1. The one-page building summary rebuilt on the advisory-20 data, Monday morning. Wind from the NHC wind structure on the NHC line scaled to each building; rain from the GOES-18 estimate carried along the line. The advisory-20 line moves away from the city from 6 am onward, so the "pass" the graphic names is simply the first hour of its window, not an approach; the real closest pass was at midnight (section 3). The window it shows is the modelled peak of a wind that the airport has not registered at all.

3. What was forecast, advisory by advisory, and what happened

The NHC issued twenty advisories for Simon between Wednesday morning and Monday morning. Figure 2 draws the first 72 hours of every forecast line over the real track; Figure 3 reduces each line to the one number a resident of Vallarta cares about, how close it would come, and the intensity at that point; Figure 4 is the NHC's own probability of damaging wind in the city, advisory by advisory.

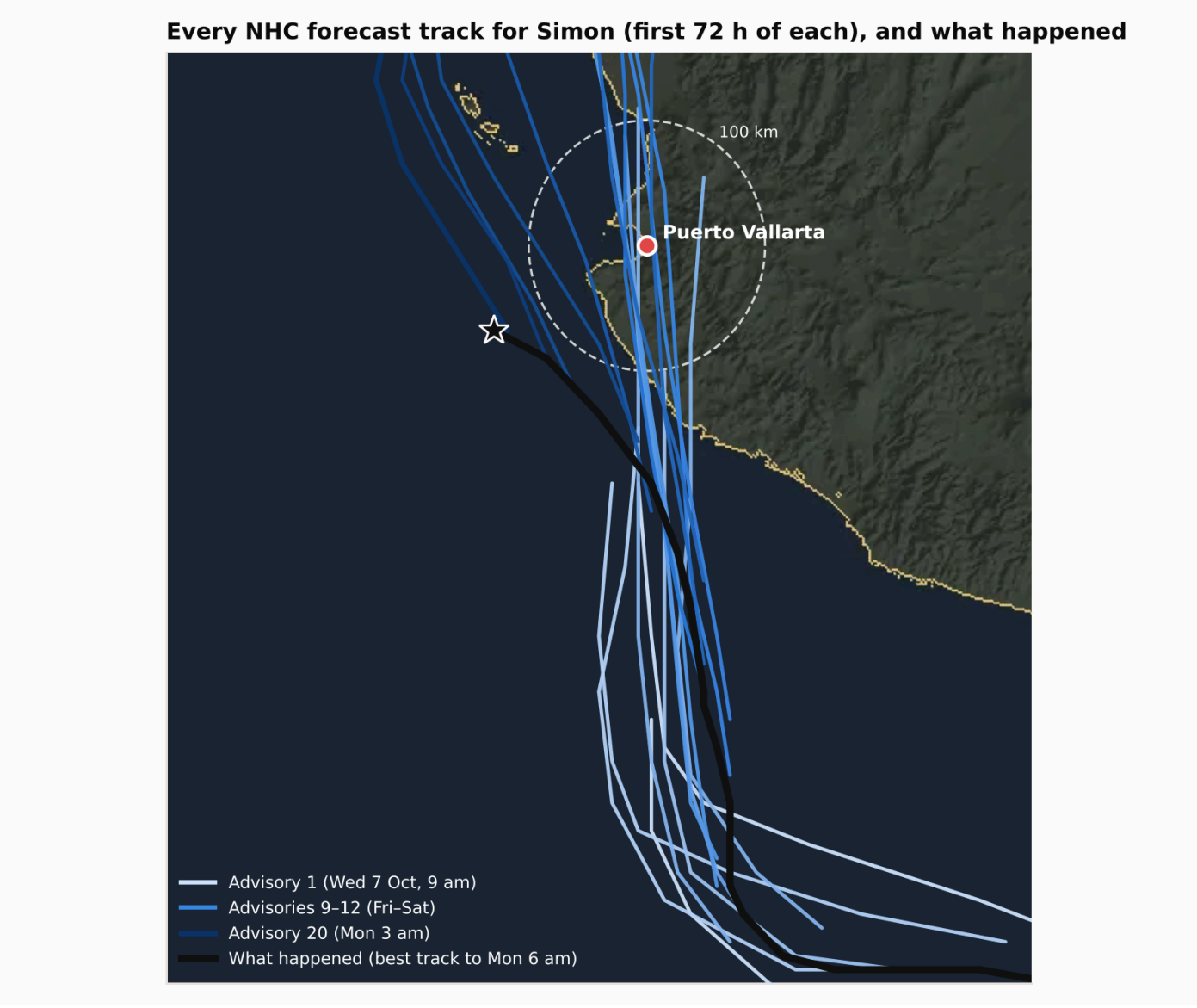


Figure 2. Every NHC forecast line for Simon (first 72 hours of each, light blue for the earliest advisory to dark blue for the latest) and what happened (black, the NHC best track to Monday 6 am). The dashed circle is 100 km around the city. The star is the position at Monday 6 am.

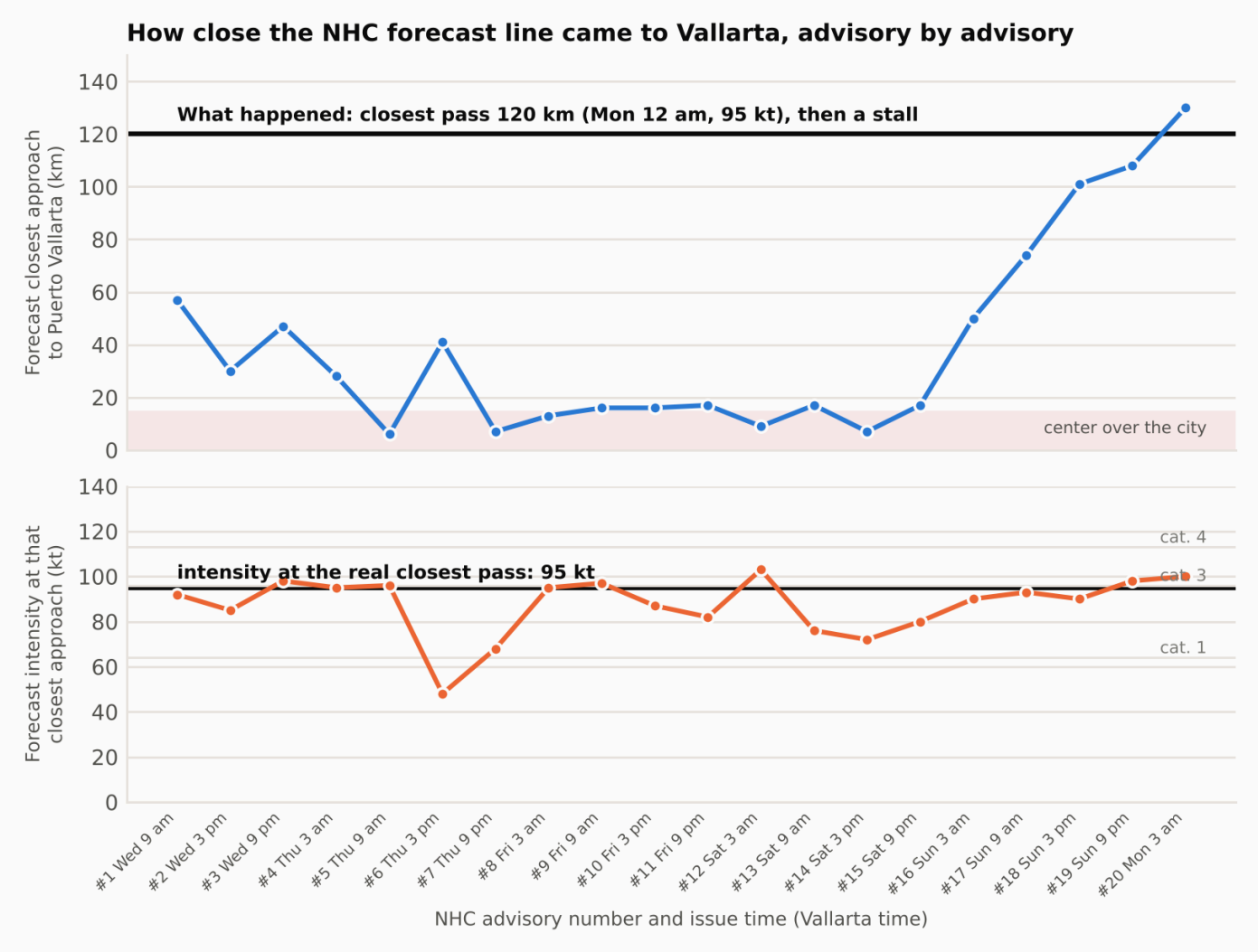


Figure 3. Top: the closest approach to Puerto Vallarta on each advisory's forecast line (km). The pink band is the width of the city. The black line is what happened: closest pass 120 km at midnight Sunday, as a 95-kt hurricane, then a stall. Bottom: the forecast intensity at that closest approach.

The sequence has three phases. **Wednesday to Saturday night (advisories 1–15):** the line came within 6–57 km of the city, within 20 km on ten of those fifteen advisories, with 48–103 kt at the closest point (68–103 kt leaving out advisory 6); the cone, drawn around that line, covered the whole bay. **Sunday morning (advisories 16–17):** as the storm held a north-northwest heading parallel to the coast instead of bending north-northeast, the line moved to 50 and then 74 km. **Sunday 3 pm (advisory 18):** 101 km, and the tropical-storm probability fell from 56% to 8% in one step. **What happened:** 120 km, farther than any advisory had forecast, and then a stall that kept the center 120–140 km from the city through Monday morning.

Two things are worth saying plainly. First, the NHC's forecast was wrong in the direction that mattered for four days, and it was wrong because the regional hurricane models it relies on for structure were wrong in that direction (section 6). Second, its probabilities were more honest than its line: a 56–76% chance of tropical-storm-force wind is not a promise, and a reader who had the probabilities would have expected an event forecast at those odds to miss one time in three or four. It missed. The line and the cone offered no way to see that.

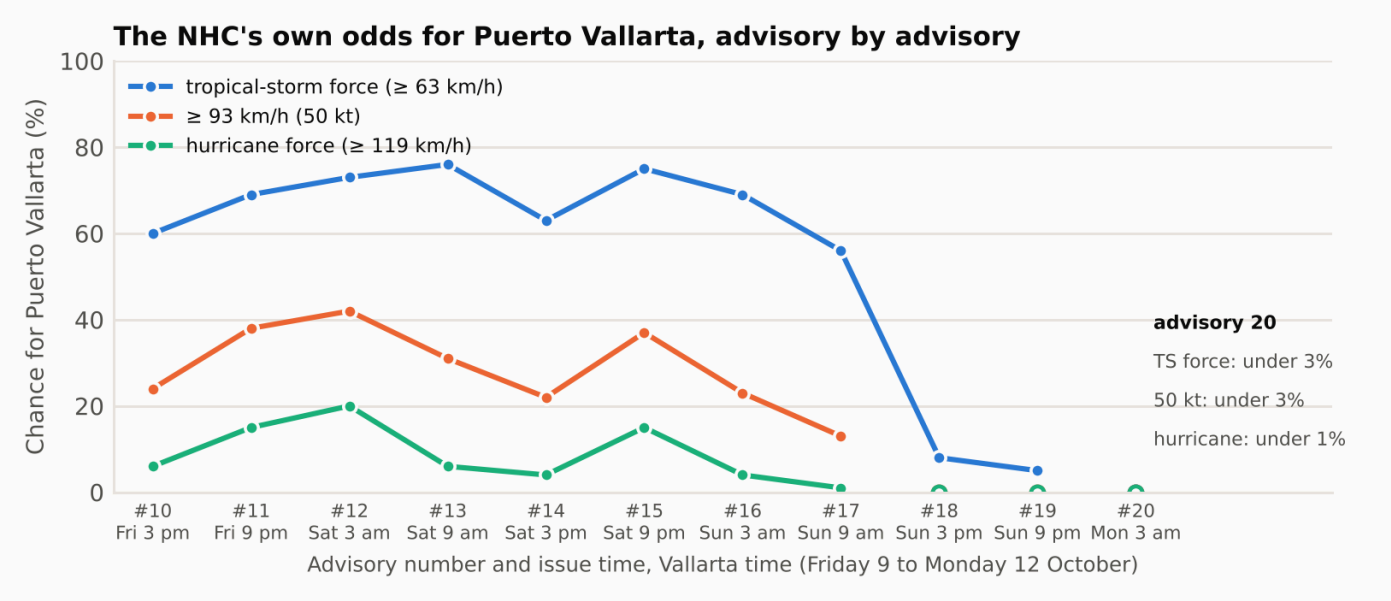


Figure 4. The NHC's cumulative probability of tropical-storm-force, 50-kt and hurricane-force wind at Puerto Vallarta, advisories 10 to 20 (the city first appears in the table at advisory 10). Open markers at zero: the location was not listed for that threshold (below 3%, or 1% for hurricane force). These numbers were always available in the wind-speed probability product; the public graphics showed the cone instead.

4. The resident graphics against the outcome

Three versions of the resident graphics were produced on Sunday (Appendix A has every page of each, in English and on a white background for this report): the first set at 9:10 am on advisory 17, a corrected set at 1 pm after a resident reported that it was not raining, and the 3 pm set on advisory 18. A fourth set, on advisory 20, is Figure 1 and Appendix A4. Figure 5 reduces the four to what they said for the three buildings and sets that against the airport record.

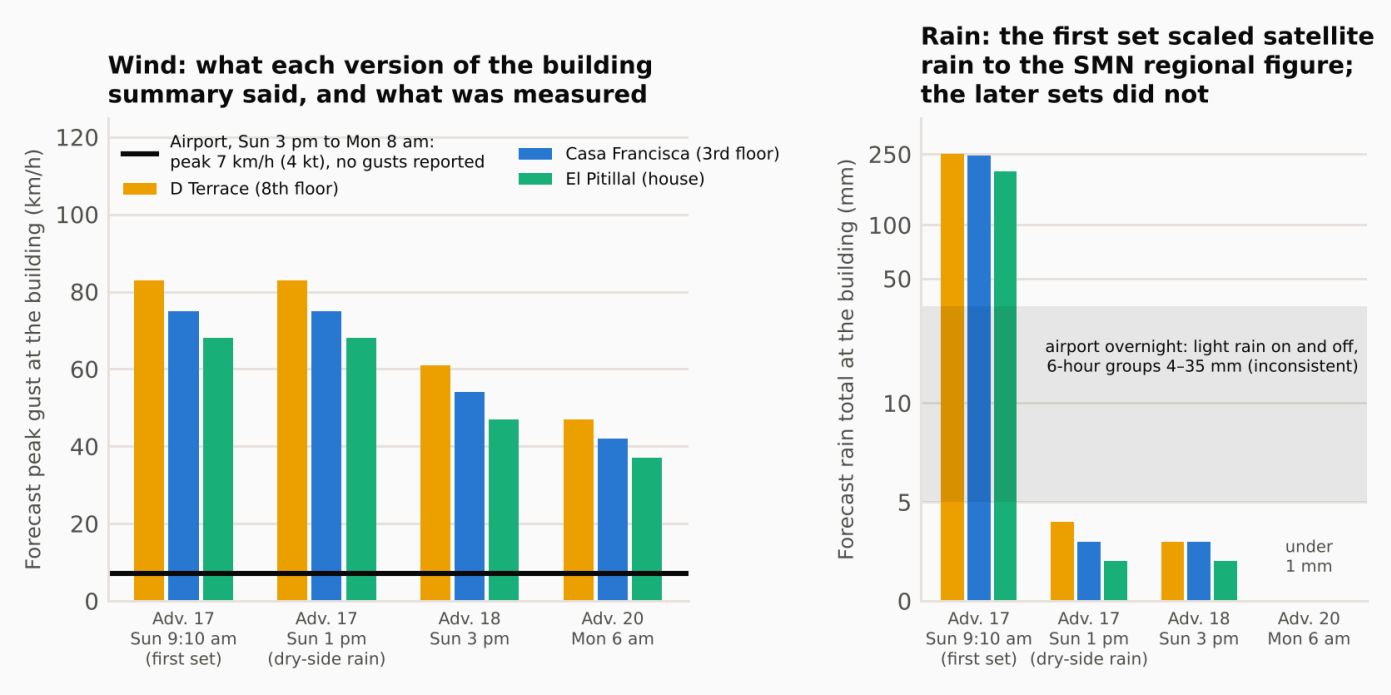


Figure 5. Left: the peak gust each version of the building summary gave D Terrace, Casa Francisca and El Pitillal, and the airport's record from Sunday 3 pm to Monday 8 am (peak 7 km/h; no gust reported). Right: the rain total each version gave; the first set scaled the satellite rain estimate to the SMN regional maximum.

What went wrong, and why. The two Sunday-morning sets did not take their wind from the NHC line at all. They took it from the NHC's probabilities: a 56% chance of tropical-storm-force wind was converted into the wind the city had a 50% chance of reaching (36 kt sustained), and that was scaled to each building by its exposure (0.70 to 0.89); the line-and-radii field gave more (42 kt) and was capped by it. In other words, the morning numbers were the NHC's own odds, read as a median. The 3 pm set was the first in which the line and the radii set the number (26 kt sustained, below the probability cap), and it was still too high, because the official 34-kt radius in the north-east quadrant was 130–150 km on Sunday morning and 93 km at 3 pm, so any line passing within that distance put tropical-storm-force gusts over the bay, and the radius is an open-water, all-quadrant envelope that says nothing about the lee of the Sierra Madre. The line itself was 45–115 km too close until Sunday afternoon. The rain error was simpler and worse: the first set multiplied the satellite estimate by 5.8 to match the SMN's "more than 250 mm", a regional maximum for "Jalisco (coast and west)" that in practice describes the windward terrain rather than the city (our reading, section 5). It was corrected within four hours because one resident said it was not raining, which is the most useful observation this storm produced. A third error surfaced on Monday morning and is worth recording because it is exactly the kind a verification step catches: the satellite centre-finder, which supplies the last four hours of observed positions to the bay fields, locked onto cold cloud tops over the Sierra and placed Simon's centre near Guadalajara; the first Monday build carried that error (winds from the

north-west) until the fixes were checked against the NHC position and replaced. Figure 1 and Appendix A4 are the corrected set.

What went right. The model scorecard (section 6) identified on Sunday morning which models to trust and which not, and the forecasts it rated best kept the center 100 km or more from the city; the satellite rain estimate, left unscaled, gave the city under 10 mm, consistent with the airport's light rain on and off (its precipitation groups are inconsistent, 4–35 mm per six hours, and there is no reliable gauge total); the timing of the closest pass (around midnight) was right; and the sea warnings were right. The graphics also carried the NHC's probabilities on every page; from 3 pm those said "unlikely" (8%) while the building numbers still said "strong", and in the morning the 56% they showed was, honestly read, a coin toss.

The lesson for the design. A resident product should lead with the probabilities and show the deterministic wind as the worst plausible case, labelled as such; it should never scale a satellite estimate to a regional forecast; and it needs a ground-truth channel, even a WhatsApp group of ten people with a view of the bay, wired into the next update. Section 10 turns that into a proposal.

5. The uploaded analysis: what it argues, and where the data agrees

The document is a long essay in five parts: the physics of tropical cyclones and of this coastline; the architecture of AI weather models and why they under-forecast intensity (training on 31-km reanalysis, a mean-squared-error loss that rewards smoothing, the "double penalty"); pathways to fix that (high-resolution training, physics-informed networks, neural ODEs, and ECMWF's AIFS-TC post-processing); a model-verification table for Simon; and a critique of the NHC cone of uncertainty as a communication device. Much of it is sound. The parts that matter for someone deciding what to do tonight are the verification table and the rainfall claim, and those are the weakest parts. The table below checks each claim against the record (as of Sunday 3 pm, advisory 18, the last full verification before the closest pass; section 3 carries the outcome) we built from the ATCF decks, the ECMWF and NOAA open-data feeds and the airport observations.

Claim in the document	What the record shows	Verdict
"Pure AI models (GraphCast, AIFS, Pangu-Weather) vastly outperformed the traditional dynamical models (GFS, ECMWF) in predicting the exact … positioning of Hurricane Simon."	Partly. Over the last 48 h, ECMWF AIFS averaged 50 km against 69 km for ECMWF IFS and 67 km for GFS. But GraphCast (66 km, incomplete) and Pangu (107 km, incomplete), run by NOAA/CIRA from GFS analyses, were no better than GFS. The best position model was DeepMind's cyclone model at 33 km, which the document never mentions, and the NHC's corrected consensus (38 km) and official forecast (47 km) beat every gridded AI model.	**Overstated**
"The dynamical models exhibited absolute supremacy regarding intensity prediction." HAFS-B "~7.8 kt" mean intensity error; HAFS-A ~9.5 kt.	No. Whole-storm mean intensity errors at 12–36 h: NHC official 5.6 kt, corrected consensus 7.7, statistical DSHP 13.0, DeepMind cyclone model 14.6, HAFS-A 18.0, HMON 20.0, HWRF 20.8, HAFS-B 20.8. From the Saturday 6 pm run, HAFS-A and HAFS-B forecast 34 and 32 kt for 6 pm Sunday, against 115 kt observed at 3 pm: they weakened it because they took it inland about 150 km too far north (Figure 8).	**Contradicted**

Claim in the document	What the record shows	Verdict
Verification table: GraphCast ~39.5 nm, AIFS ~38.0, Pangu ~41.2, HAFS-A ~48.2, HAFS-B ~51.0, ECMWF HRES ~58.4, GFS ~62.1 nm track error for "72-hour lead time forecasts (initialized on October 8, 2026)".	No source, no cycle hours and no averaging method are given. Computing that metric from the public decks (the four Oct 8 cycles, 72-h forecasts against the best track) gives HAFS-A 87 nm, HAFS-B 83 nm, HWRF 56 nm, GFS 106 nm and the NHC 70 nm for track, with intensity errors from 36 kt (NHC) to 64 kt (HAFS-B): every model, physics or AI, missed the rapid intensification at 72 h by 13 to 110 kt (36–64 kt on average for the models the document lists), so the table's 8–10 kt for the HAFS models cannot be right. In our 48-hour window HAFS-B had the largest errors of any model (169 km mean; 149 km too far north at 3 pm) and HAFS-A 117 km.	**Unsupported**
"Uncorrected AI models forecasted Simon as a mere Category 1 or strong Tropical Storm, missing the rapid intensification event entirely."	Confirmed. For 6 pm Sunday, from the Saturday 6 pm analysis (115 kt observed at 3 pm): ECMWF AIFS 34 kt, FourCastNet 13, Pangu 18, Aurora 18, AIGFS 16, GraphCast 24 kt at 18 h. The physics ensembles on their 25-km public grids did the same: the median ECMWF ensemble member had 50 kt at 24 h (members 20–106), the AIFS ensemble median 33 kt.	**Confirmed**
The cause is resolution of the training data plus the smoothing effect of the MSE loss ("double penalty").	Consistent with what we saw, and the standard explanation. One addition: for GFS, AIGFS and the CIRA runs part of the collapse came from drifting the low inland, so track and intensity errors compound; the DeepMind cyclone model, trained for cyclones specifically, kept 91 kt at 24 h.	**Reasonable**
AIFS-TC post-processing cuts the AI intensity bias "from nearly -29 knots to an operational margin of about -2 knots" and "was autonomously designed and built by a Large Language Model coding agent".	Not testable here. AIFS-TC output is not part of the ECMWF open-data feed we used, and the claims about its construction come from the cited preprint. They are reported, not verified.	**Unverified**
"When the storm's track shifts, or when a massive, highly asymmetric system like Simon brings 15 inches of rain to a mountainous area like Puerto Vallarta that was technically 'outside' the center-track cone, it breeds profound public distrust."	Used as an example, not a forecast, but it has not happened: by 3 pm Sunday residents reported no rain in the city and the airport about 9 mm. The SMN's "more than 250 mm" is for "Jalisco (coast and west)" and the NHC's "isolated maximum storm totals of 30 inches" for "portions of the states of Michoacán, Colima and Jalisco", in practice the windward terrain; with the center to its southwest, Vallarta is in descending easterly flow.	**Illustrative, overstated**
The Sierra Madre shields the city from the strongest wind; orographic lift makes flash flooding and mudslides the lethal hazard.	Agree on the mechanism. Our bay-wide fields show the same: wind reaching the city from the east and east-southeast, rain concentrated on the sierra, an onshore turn only once the center is northwest of the bay (Monday afternoon in the consensus).	**Agree**
Rapid intensification driven by deep ocean heat content and low shear.	Consistent. The best track shows 65 kt at 6 pm Thursday and 105 kt at 6 pm Friday (+40 kt in 24 h), then 135 kt by Saturday evening.	**Consistent**

Claim in the document	What the record shows	Verdict
The cone of uncertainty is misread (size, center-line and edge fallacies) and triggers anxiety; the 2026 cone embeds inland watches and warnings and experimental versions have sought to capture 90% of errors.	Agree on the substance; the three fallacies are well documented, and the inland watches-and-warnings cone is real. The specific "90%" figure and the psychometric claims could not be checked from here. See section 7 for what we did instead.	**Agree, specifics unverified**

The document's 60-item reference list is not reproduced here. Several of its entries are forum threads and news items; the technical claims that matter are checked above against primary data rather than against those sources.

6. The verification, model by model

This section is the verification as it stood on Sunday at 3 pm (advisory 18), the last full set before the closest pass. Every model was scored the same way. **Position:** the error of each forecast made in the 48 hours before 3 pm Sunday, measured against the NHC's official positions every six hours and against the advisory-18 position; the ranking score is the mean at 12, 24 and 36 h lead. **Vallarta wind and rain:** what the run of Saturday 6 pm gave the city up to 3 pm Sunday, against the airport record (max 9 kt in the available reports; about 9 mm of rain by 1:44 pm). Gridded models: 10-m wind interpolated to the city, rain at the nearest grid point; track models give no point wind, so their wind is the value implied by the NHC wind structure, or the model's own wind radii where the deck carries them, placed on the model's line.

What stands out

- **The best three were not the newest or the highest-resolution models.** DeepMind's cyclone model, the NHC's corrected consensus and the NHC official forecast. The simple climatology-and-persistence baseline (53 km) beat GFS, GEM, UKMET and all five US regional hurricane models (HWRF, HMON, HAFS-A, HAFS-B, COAMPS-TC) over this window, which is unusual and says the storm's motion was steady while many models were looking for a turn.
- **The regional hurricane models failed in the direction that mattered.** HAFS-A, HAFS-B, HMON and COAMPS-TC were 140–310 km too far north at various cycles, carrying the center into or past Banderas Bay as a weakening storm. Because these are the models the NHC leans on for structure, the official line followed them north with the NHC's own intensity, which is why the official forecast was 60–140 km too far east and north on Friday and Saturday, and why the NHC's own probability of hurricane-force wind in Vallarta reached 15–20% late Friday and early Saturday (section 3).
- **Among gridded AI models, ECMWF AIFS and its ensemble did well on position** (50 and 58 km; 43 of 51 ensemble members within 100 km of the true center at 3 pm). The GFS-initialised models in the NOAA/CIRA archive did not: FourCastNet and Aurora were 160–200 km off at 3 pm, and the NOAA AIGFS dissipated the storm inland and ended 210 km too far north.
- **Local wind and rain are beyond all of them.** Nine gridded models put the city's peak wind at 4–12 km/h against 17 km/h measured, and five of the seven that carry rain gave more than fell. The bay, the city and the sierra do not exist on a 25–50 km grid; the NBM, the US blend that corrects for terrain, reaches Vallarta only through its oceanic domain (wind only); its 36 km/h was above the airport's 17 km/h, as were the track-based estimates (28–70 km/h), while the coarse grids were below it.

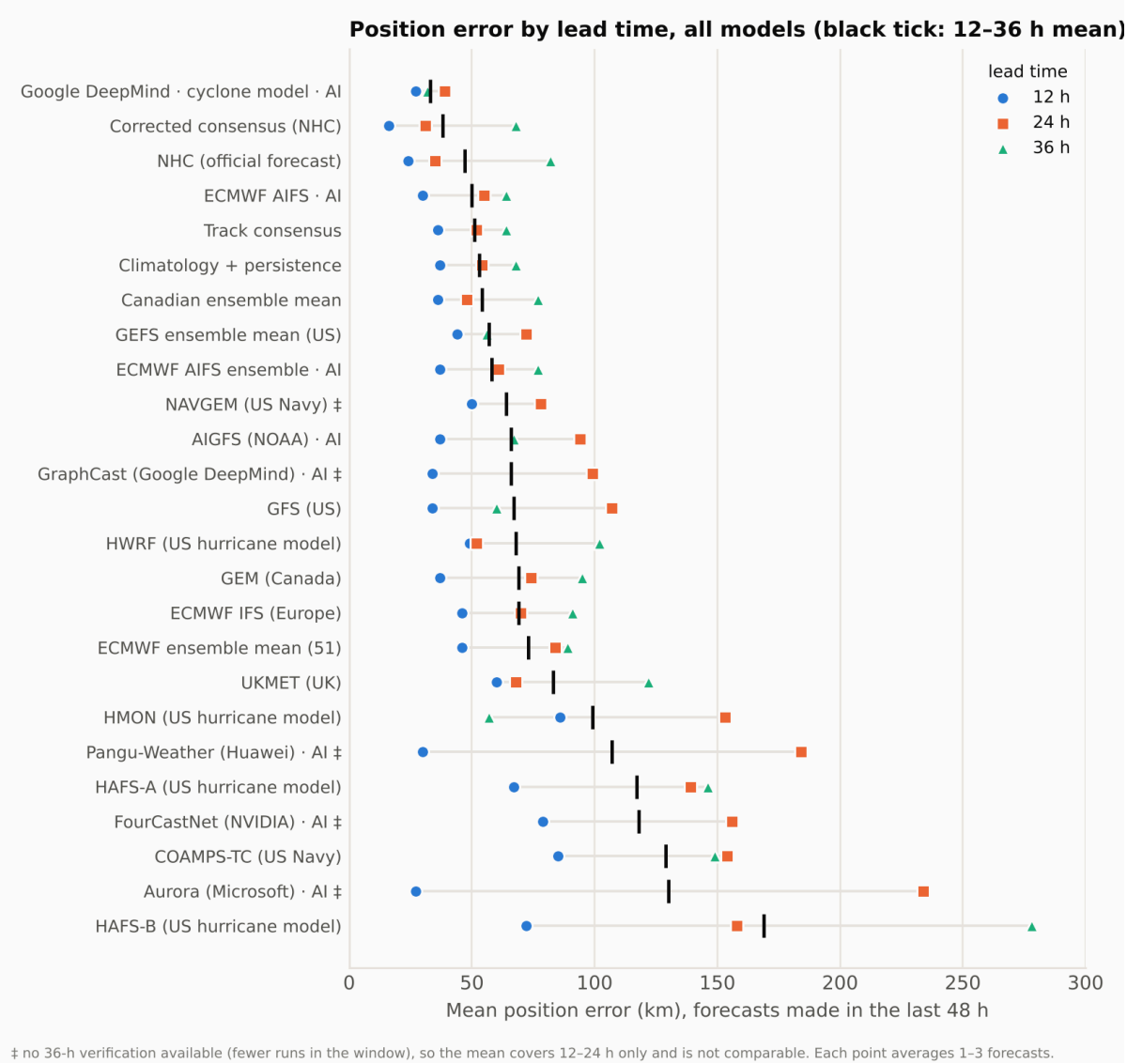
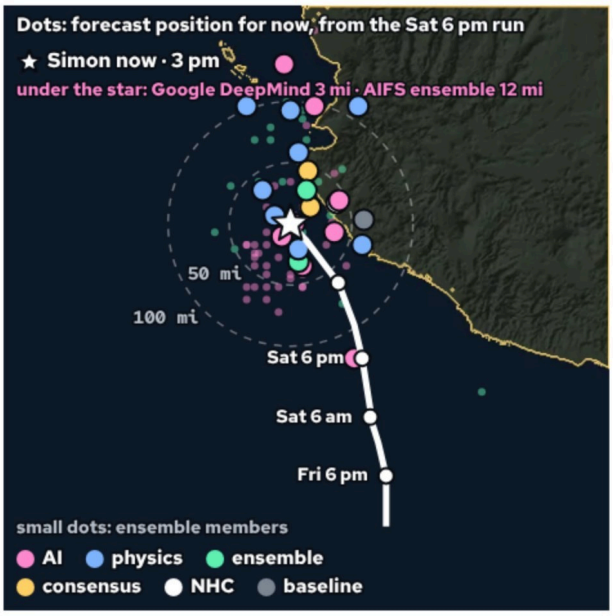


Figure 6. Position error by lead time. The black tick is the 12–36 h mean used to rank the models. Models marked ‡ lack a 36-h verification inside the window, so their score covers 12–24 h only and is not directly comparable. Each point averages one to six forecasts (six at 12 h and four at 24 h for the 6-hourly models; one to three at 36 h and for the 12-hourly ones): a small sample, so the order of neighbours can change, but the gap between the top three and the bottom five is far larger than the noise.

SIMON · 6/6 NHC advisory 18 · Sun Oct 11, 3 pm · Vallarta time

Which model got it right? The last 2 days, checked

Each model against what happened: where it put Simon (versus the NHC's official position) and the wind and rain it gave Vallarta, versus the airport record. Reference run: Sat 6 pm, checked today at 3 pm.



Position · mean error at 12–36 h, last 2 days
1. Google DeepMind (21 mi) **2.** Corrected consensus (24 mi) **3.** NHC (29 mi). Best physics model (fully verified): GFS (42 mi); best ensemble: ECMWF ensemble mean (34 mi).

Right now (3 pm) · what the Sat 6 pm run said
Closest: Google DeepMind 3 mi · ECMWF AIFS ensemble 12 mi · ECMWF IFS 15 mi. **Farthest:** AIGFS 131 mi N · FourCastNet 122 mi SSE · UKMET 111 mi NNE. Ensemble members within 62 mi (100 km) of the real center: GEFS 11 of 20; ECMWF AIFS ensemble 43 of 51.

Vallarta wind · forecast versus measured
Measured: max wind 13 kt (15 mph) at the airport (METAR, 48 h). Closest gridded model: **HAFS-A** (16 mph); 7 gridded models (GFS, GraphCast, FourCastNet…) gave 5–8 mph (too low: the bay does not fit a 15-mile grid). With its own wind structure, DeepMind implied 29 mph and over open water off the city; track-based models as a group implied 25–46 mph: too high.

Vallarta rain · forecast versus measured
Measured: no rain until 1 pm, then one shower (~0.35 in at the airport). Forecast for the same period: AIGFS 0.4 in · GFS 0.7 in · GEFS ensemble mean 0.7 in · HAFS-A 1.0 in · GraphCast 1.5 in. AIGFS got it right; the rest were too wet, GraphCast by a lot. DeepMind's cyclone model gives no rain (position, intensity and wind radii only); its gridded model is GraphCast.

Every model, one by one

Figure 7a. The verification graphic shared with residents (English; Spanish on the shared page), upper part. Map: where each Saturday-evening run put Simon for 3 pm Sunday, with the ensemble members as small dots.

	Mean error 12–36 h · 2 days	Where it put it today Sat 6 pm run	Max PV wind measured 15 mph	PV rain measured 0–0.4 in
Google DeepMind · AI	21 mi	3 mi E	29 mph*	–
Corrected consensus (NHC)	24 mi	21 mi NE	38 mph*	–
NHC (official forecast)	29 mi	41 mi ENE	44 mph*	–
ECMWF AIFS · AI	31 mi	35 mi SSE	–	–
Track consensus	32 mi	45 mi NNE	–	–
Climatology + persistence	33 mi	60 mi E	–	–
ECMWF ensemble mean	34 mi	32 mi SSE	28 mph*	–
GEFS ensemble mean (US)	35 mi	30 mi NNE	6 mph	0.7 in
ECMWF AIFS ensemble · AI	36 mi	12 mi SW	–	–
NAVGEM (US Navy)	40 mi‡	61 mi ESE	25 mph*	–
AIGFS (NOAA) · AI	41 mi	131 mi N	5 mph	0.4 in
GraphCast (Google) · AI	41 mi‡	44 mi ENE	7 mph	1.5 in
GFS (US)	42 mi	59 mi N	8 mph	0.7 in
HWRF (US hurricane model)	42 mi	35 mi NW	39 mph*	–
GEM (Canada)	43 mi	22 mi SSE	26 mph*	–
ECMWF IFS (Europe)	43 mi	15 mi WNW	–	–
UKMET (UK)	52 mi	111 mi NNE	39 mph*	–
HMON (US hurricane model)	62 mi	–	43 mph*	–
Pangu-Weather (Huawei) · AI	67 mi‡	37 mi E	6 mph	–
HAFS-A (US hurricane model)	73 mi	103 mi NNW	16 mph	1.0 in
FourCastNet (NVIDIA) · AI	73 mi‡	122 mi SSE	7 mph	–
COAMPS-TC (US Navy)	80 mi	–	46 mph*	–
Aurora (Microsoft) · AI	81 mi‡	98 mi NNE	6 mph	–
HAFS-B (US hurricane model)	105 mi	93 mi N	46 mph*	–
NBM · National Blend (US)	–	–	22 mph	–

Reading it: over the last two days Google DeepMind (artificial intelligence) was the most accurate on Simon's position: mean error 21 mi, and its Sat 6 pm run put the center 3 mi from where it is now. Next come Corrected consensus and NHC. HAFS-B, Aurora, HAFS-A, UKMET and AIGFS carried it 93–131 mi farther north, already over or past the bay; among them US hurricane models, which is where the hurricane-wind scenario for Vallarta came from. Locally only AIGFS got the rain right: the rest gave Vallarta more rain than fell, and almost all less wind. The sierra and the bay do not fit 15–30 mile grids.

For what is ahead, follow Google DeepMind, the NHC consensus and the ensemble means, not the ones that missed. Today they put the closest pass about 65 mi southwest of Vallarta around 11:30 pm Sunday, with a peak sustained wind of 35 mph over open water off the city (less on land); for the next 24 h the gridded models give Vallarta 0.1 in (GFS) to 0.9 in (AIGFS); GraphCast goes to 4 in, but it has been the wettest miss.

‡ Score without the 36-h lead (too few runs): not comparable with the others. † Implied wind using the model's own wind radii on its line; * using the NHC radii (track models give no point wind). Grid: wind interpolated to the city and rain at the nearest point, on 15–30 mile grids; rain accumulated from the run to 3 pm. Position: against the NHC's official track (6-hourly) and advisory 18. Sources: NHC/ATCF, ECMWF open data, NOAA (GFS, GEFS, AIGFS, NBM, HAFS), the NOAA/CIRA archive of AI models run from GFS analyses (FourCastNet, Pangu, GraphCast, Aurora), METAR MMPR.

Not an official forecast. Follow the SMN, the NHC and Protección Civil. Emergencies: 911.

Figure 7b. The same graphic, lower part: all 25 models, with the reading and the notes.

7. The intensity problem, shown on Simon

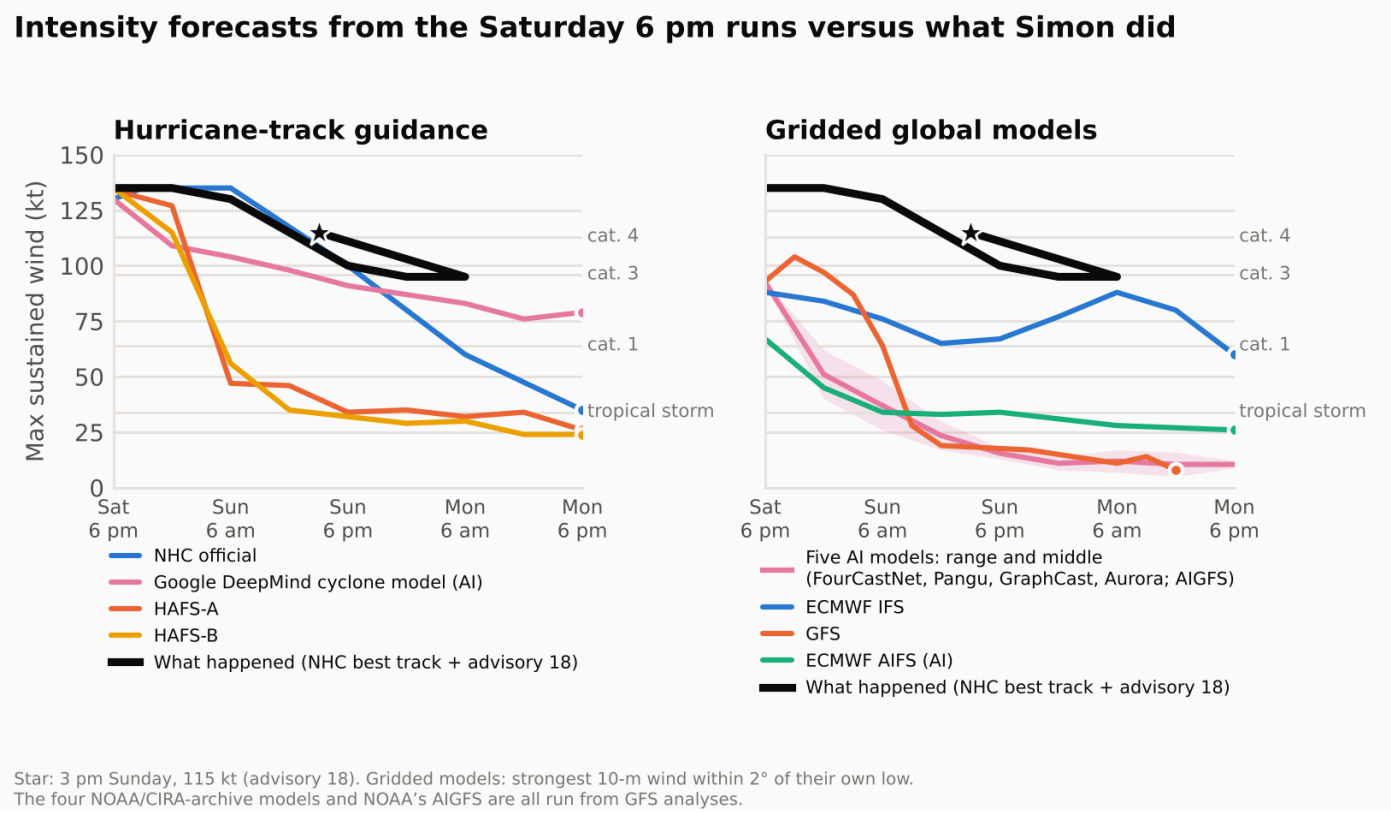


Figure 8. Intensity forecasts from the Saturday 6 pm runs. Left: the guidance the NHC uses. Right: gridded global models, with the four AI models from the NOAA/CIRA archive and NOAA's AIGFS as a band. The black line is what Simon did; the star is advisory 18.

Three different failures are visible in one picture. The **gridded AI models** (right, pink band and the AIFS line) start from the same 93-kt GFS or 67-kt AIFS analysis and lose half their wind in the first six hours, then settle at 5–35 kt: the smoothing the document describes, plus a 0.25° grid that cannot hold a 20-km eyewall. The **regional hurricane models** (left, HAFS-A and HAFS-B) start at the right intensity and crash to about 35 kt by 18–24 h, not because of smoothing but because they ran the storm into the Sierra Madre 150 km north of where it went. The **global physics models** on their 25-km public grids (ECMWF IFS, GFS) sit in between: IFS kept 60–88 kt, GFS fell to 19 kt as it too took the center inland.

The guidance that worked was the guidance that treats intensity as its own problem: the NHC forecaster (135 kt at 12 h, 100 kt at 24 h), the corrected consensus and the DeepMind cyclone model (104 kt at 12 h, 91 kt at 24 h), which is trained on cyclone tracks and intensities rather than on gridded fields. That supports one half of the document's prescription, a cyclone-specific correction on top of the global model (what AIFS-TC and the DeepMind model are), and does not test the other half, physics-informed networks.

Two caveats. The intensity verification covers one storm and a handful of cycles. And "right" here means right for Vallarta's purposes, where the question was whether a category-4 core would reach the bay; for the coast between Playa Pérula and Cabo Corrientes, which the eastern eyewall is brushing this evening, the intensity the NHC carried was the relevant number, and it was right.

8. Uncertainty, shown rather than hidden

This section is kept as it was written on Sunday at 3 pm, before the closest pass, because that is when it was needed; the outcome (section 3) is the test of it. The document's criticism of the cone is that it looks certain. The honest alternative is to show how the forecast itself moved (Figure 4), how wide the ensembles are, and what the models' own error record implies.

Figure 4 (section 3) shows the NHC's own odds. From Friday afternoon to Saturday night (advisories 10 to 15) the chance of hurricane-force wind in the city moved between 4% and 20%, and the chance of tropical-storm-force wind between 60% and 76%. That is the period in which the regional hurricane models carried the core into the bay. Since then it has gone 75%, 69%, 56% and, at 3 pm today, 8%: almost the whole drop came in one advisory, between 9 am and 3 pm, as the storm held its north-northwest heading and the guidance shifted west. A number that can fall 48 points in six hours can also climb back. Anyone who read the probabilities rather than the cone saw a "likely tropical-storm wind, unlikely hurricane" picture on Saturday and a "probably neither" picture today. The cone, drawn around the official line that was 60–140 km too far east, looked the same both days.

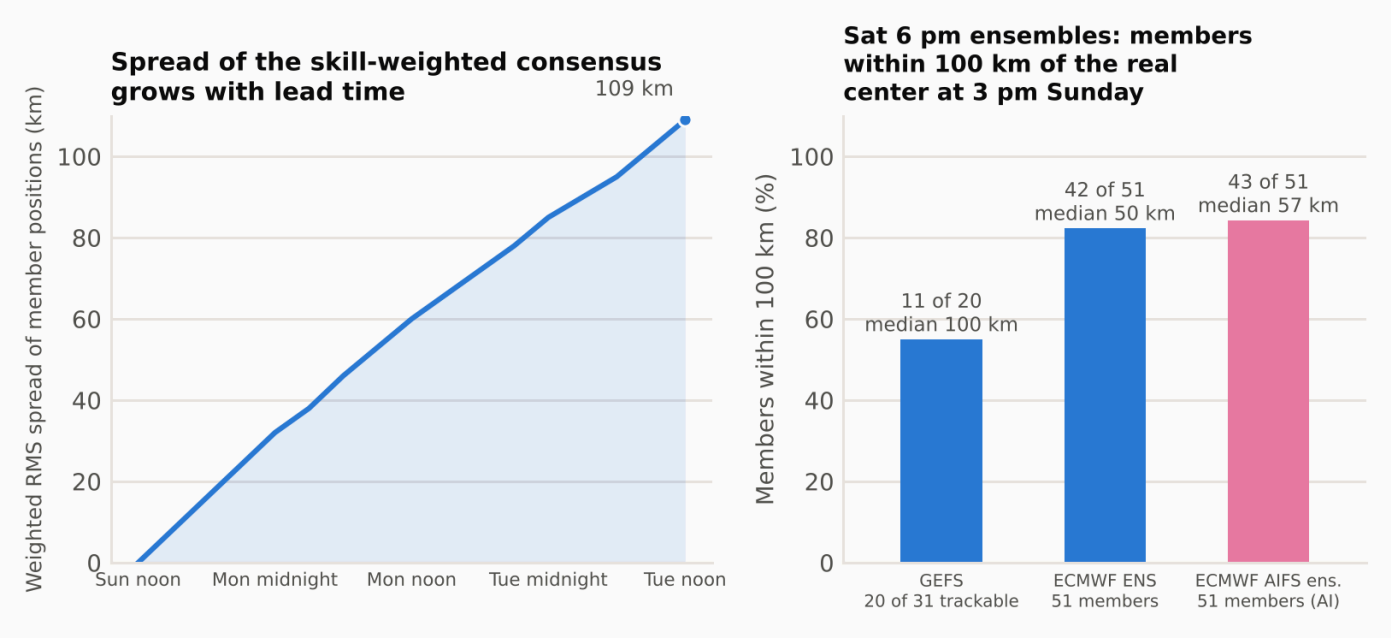


Figure 9. Left: the spread of the skill-weighted consensus (eleven models weighted by their 48-hour record) grows from zero to about 80 km at 48 h. Right: of the ensembles run Saturday evening, the share of members that had Simon within 100 km of where it was at 3 pm Sunday.

101–109 km
closest pass to the city, middle half of the weighted models (median 101 km, west-southwest, about 11:30 pm)

0%
of the weighted models bring the center within 50 km of the city; 10% within 100 km

54–58 km/h
peak sustained wind off the city over open water, middle half of the weighted models (median 31 kt); 7% of the weight reaches tropical-storm force

8%
the NHC's own odds of tropical-storm-force wind at Puerto Vallarta (advisory 18); under 3% for 50 kt, under 1% for hurricane force

What could still go wrong

- **An eastward wobble.** The center is 90 km south of Cabo Corrientes. The NHC's 34-kt wind radius in the northeast quadrant is about 93 km. A 30–40 km shift east of the forecast line, inside the two-day average

error of most models and at the edge of the best three's (33–47 km), would bring tropical-storm-force gusts over the bay tonight, with sustained winds just short of it. Treat the 8% as "unlikely", not "impossible".

- **Rain when the wind turns.** While the center is southwest of the city the flow descends the sierra and the city stays dry, as it did today. Once the center passes to the northwest (the consensus says Monday afternoon) the wind turns onshore and showers become likely; the gridded models give 3–24 mm for the next 24 h, GraphCast 106 mm, but GraphCast has been the wettest miss of all. The SMN's extraordinary totals apply to the coast and sierra south of the bay, where the Cuale and Pitillal rivers rise; river levels, not local rain, are what to watch in the lower town.
- **The sea.** This is the one hazard every source agrees on: waves of 8 to 10 m on the outer coast, a surge of 1.5 to 2.5 m, water pushed into the bay mouth, dangerous conditions on the malecón and the southern beaches through Monday.
- **Our own estimates.** The city wind numbers are the NHC wind field on the NHC line, capped by the wind value the NHC odds give a 50% chance of, and scaled by building exposure; they are not a measurement and they move with every advisory. The rain field is a satellite estimate carried along the NHC line, and there is no radar in the region. The verification itself rests on one to six forecasts per model and lead.

9. Risk communication: my own input

Where I agree with the document. The cone communicates the wrong thing to a resident. It answers "where might the center be" with a line and a two-thirds envelope, and people read it as the storm's size and as a yes-or-no about their town. For Vallarta the right question was never where the center would be; it was "what wind reaches my building, when, and what rain reaches the rivers above me". The NHC does publish the answer to the first part, as wind-speed probabilities and as earliest-arrival times, and this weekend those numbers told a calmer and more accurate story than the cone did. The 2026 change that draws inland watches and warnings inside the cone helps, but as long as a single black line is on the graphic, people will anchor on it.

Where I would push back. The document treats AI versus physics as the main axis. On this storm the useful axis was "models that treat the cyclone as its own object" (the NHC forecaster, the consensus aids, DeepMind's cyclone model, and in the document's own account AIFS-TC) versus "models that treat it as a feature on a global grid" (every gridded AI model and the 25-km ensembles). And the table at the center of its verification section is not a verification; it is a set of plausible-looking numbers without a source. In a document meant to be forwarded during a hurricane, that matters more than the quality of the physics chapters.

What we did instead, and its limits. The six numbered graphics on the shared page (Appendix A here) and the one-page building summary (Figure 1) replace the cone with: the NHC's probabilities turned into a wind number per building and floor; a timeline of when it starts, peaks and ends; a map of where water and slopes are the danger; and an explicit scorecard of which models to believe, updated each advisory. The limits are real: no radar, 25–50 km model grids that cannot see the bay, wind estimates that lean on the NHC's probability product, and a verification sample of two days. The scorecard is also only as good as the next advisory; it is rebuilt at each one.

Three recommendations for the next storm. Lead with the wind-speed probabilities and the earliest-arrival time for named places, not the cone. Show the ensemble members as dots, as in Figure 7, so the spread is visible. And publish a running scorecard of model performance for the storm in progress; it would have taken the hurricane-model scenario off the table a day earlier this time.

10. Training, collaboration and a public-private partnership: what this coast needs

This section steps back from Simon. The storm passed Puerto Vallarta by, and the city was lucky rather than well served: for four days every product a resident could see, including ours, pointed the center at the bay. The global models were not the weak link. The weak link was everything between a 28-km global grid and a window in Amapas, and that gap is local, fixable, and will matter more as the storms on this coast change. What follows is my own reading of what is needed and how a realistic partnership could pay for it. Where a number is my estimate rather than a sourced figure, it says so.

10.1 What Simon exposed

Layer	What existed during Simon	What was missing	Who could supply it
Observations over the bay	One airport station (METAR, with a 19-hour gap on Saturday night); GOES-18; scattered SMN stations; no radar within range; aircraft fixes at the NHC's discretion.	Radar coverage of Banderas Bay; a dense coastal mesonet (wind, rain, pressure) at the hotels, the marina, the hills; buoys and gliders off Cabo Corrientes; a reliable ground-truth channel from residents.	SMN and the Apixqui buoy programme; hotels, the port and the airport as station hosts; a university lab as operator; residents.
Models	Global physics (ECMWF, GFS, UKMET, GEM), regional hurricane models (HAFS, HWRF, HMON), AI models (DeepMind, AIFS, AIGFS, GraphCast and others), ensembles; all open or semi-open.	Nothing between those models and the city: no local wind downscaling that knows the Sierra Madre, no bias correction trained on local observations, no local rain model.	A small technical team (university plus one private AI lab) working on open model output.
Verification	Done here, by hand, from the ATCF decks and open archives; published nowhere official until after the storm.	A public, running scorecard: which model has been right over the last 48 hours, for this storm, for this city.	The same team; the SMN as publisher.
Translation into local risk	NHC cone and probabilities; SMN regional avisos; the SIAT-CT colour; our building-level graphics (with the errors section 4 describes).	Official building-scale, probabilistic, bilingual products: "40% chance of gusts over 80 km/h at your address between 6 pm and 2 am".	CENAPRED and the state civil protection unit as owners; a private design partner; the hotel association as distributor.
People	SMN forecasters; the NHC's two-week regional workshop; the state and municipal civil protection units; academics consulted by the press.	Forecasters trained to use AI guidance and to say what it cannot do; a local duty roster during storms; feedback from the field into the next update.	NHC and WMO training; the university; the partnership as the roster's home.

10.2 Not a one-storm problem

Simon is the latest in a list of eight hurricanes since 2002, drawn in Figure 10 from the NHC's own reports, that strengthened quickly within a day or two of Mexico's Pacific coast; it is not an exhaustive list. Kenna (2002) went from 40 to 140 kt in 42 hours and hit San Blas; Patricia (2015) gained 105 kt in 24 hours on its way to the Jalisco coast; Lidia (2023) gained 45 kt in 18 hours and made landfall 35 nautical miles south-west of Puerto Vallarta as a category 4; Otis (2023) gained 90 kt in 21 hours and destroyed Acapulco; John (2024) went from a 35-kt

tropical storm to a category 3 in 24 hours and was not forecast to be a hurricane in any of the three advisories issued in the 30 hours before landfall. The mean official intensity errors over those storms ran from 6 kt (Lidia) to 35 kt (Patricia) at 24 hours and from 7 to 58 kt at 48 hours; in Kenna, Patricia and Otis they were 27–35 kt and 52–58 kt, between one and three Saffir-Simpson categories, and the NHC wrote after Patricia and again after Otis that nothing in its guidance had anticipated the strengthening. Simon, like Roslyn and Erick, had its rapid intensification forecast in advance (the advisory on Thursday night called for 100 kt in 24 hours and got 105), and even then the 48-hour value was 25 kt low.

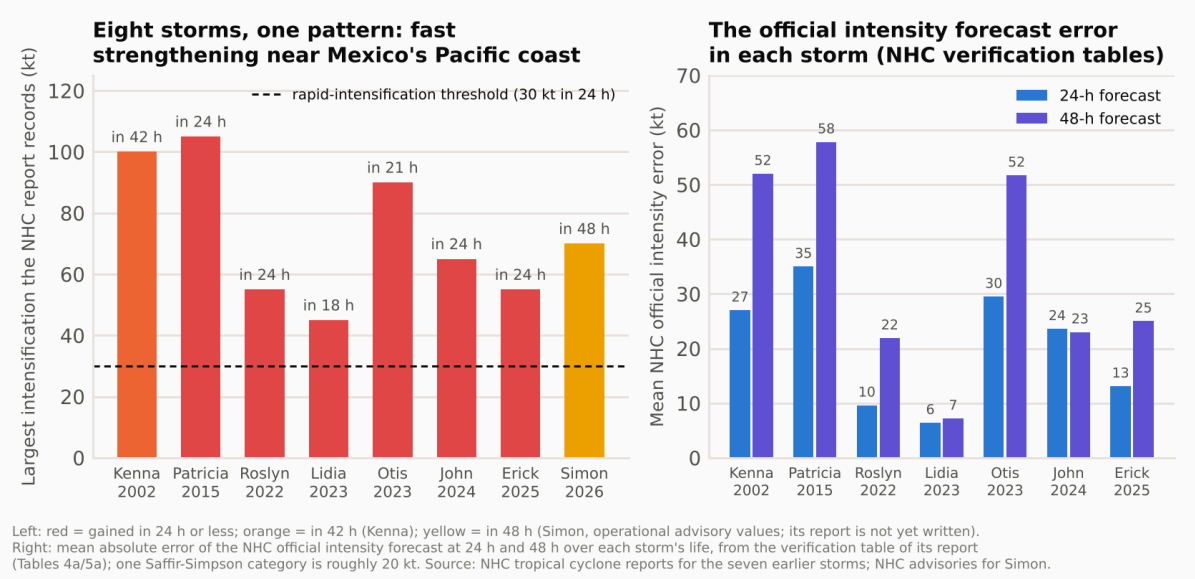


Figure 10. Left: the largest intensification the NHC tropical cyclone report records for each storm and how long it took; all eight exceed the rapid-intensification threshold. Right: the mean official intensity error at 24 and 48 hours from each report's verification table. Simon's values are operational and will change when its report is written.

What the climate adds. The IPCC's sixth assessment rates it likely that the global share of category 3–5 storms and the frequency of rapid intensification have increased over the last 40 years, and very likely that peak winds and the share of category 4–5 storms will rise with warming; its central estimates are a 10–13% increase in the category 4–5 share and an 11–14% increase in storm rain rates at 1.5–2 °C. Two findings bear directly on this coast. Balaguru and colleagues (2024) find that intensification rates close to coasts have risen by about 3 kt per 24 hours since 1979 and are projected to keep rising as wind shear near coastlines weakens. Radfar and colleagues (2026) find the clearest upward trend in storms that rapidly intensify during marine heatwaves in the eastern Pacific and the North Atlantic, and name Otis as the example. The Pacific tide gauges give a 25-year storm-driven water-level excess of about 0.85–0.95 m today, on top of a sea level that is extremely likely to keep rising. Mexico's president said in June that the Pacific is the least-observed part of the country's weather network. None of this says the next storm will hit Vallarta. It says the next storm that threatens it will more often strengthen late, close in, where today there is nothing to measure it with.

10.3 What the models can and cannot do, and what "AI plus physics" should mean

The verification in this report (sections 6 and 7) matches what the NHC found over a full season. In 2025, the first year the NHC used AI models in operations, Google DeepMind's cyclone ensemble beat the official forecast and every consensus aid on track at 48–120 hours in the eastern Pacific, and it was the only model to beat the official track at nearly every lead time for Priscilla, the storm that ran up this coast a year ago. On intensity the same model was best only out to 48 hours; beyond 60 hours the physics-based regional models (HAFS) were still the best guidance. The gridded AI models, trained on reanalysis that smooths hurricane cores, under-forecast intensity badly: the first NHC-style evaluation (DeMaria and colleagues, 2024) found their intensity errors worse

than climatology, and ECMWF's own analysis of its AI model over 70 storms in 2025 found a mean bias of −28.8 kt, with errors of 71 kt in rapid-intensification cases. Simon reproduced all of this: DeepMind's cyclone model had the best positions (33 km mean error) and sensible intensities (104 kt when the storm was 130 kt); every gridded AI model collapsed Simon to a tropical storm; the regional physics models had the intensity but carried the center 150 km too far north.

The most instructive result of the year is how ECMWF fixed its intensity problem. A post-processor trained on nine seasons of past forecasts and observations (2016–2024), using gradient-boosted trees and a small convolutional network on the storm-centred fields, cut the AI model's intensity error from 28.9 to 10.8 kt, level with DeepMind's model and with the NHC's official forecast; the authors note the system was built in hours. The lesson is not that AI is magic. It is that the cheapest, fastest-improving layer in the whole chain is a statistical correction trained on data, and that the layer is only as good as the observations it is trained on. For track and intensity those observations exist (best tracks, aircraft, satellite). For wind and rain in Puerto Vallarta they do not: there is no archive of what the bay actually does when a hurricane passes 100 km offshore, because nothing has been measuring it.

So "AI plus physics" for a coast like this should mean four specific things:

- **Physics for structure and intensity, AI ensembles for track and probability.** Use the regional hurricane models for the wind field and the intensity, the AI cyclone ensembles (50–64 members, minutes to run) for where the center goes and how uncertain that is, and show the disagreement rather than averaging it away.
- **A local layer trained on local observations.** A downscaling and bias-correction model for Banderas Bay, trained on the mesonet and radar the partnership installs, that turns a global forecast into wind and rain at the hotel strip, the marina, the river mouths and the hillsides. This is a small model and a small team; the data is the hard part.
- **Verification as a product, not an afterthought.** A running scorecard like section 6, published by the SMN during the storm, so that forecasters and the public can see which guidance has been right for this storm and weight it accordingly. NOAA's own public page on AI hurricane forecasting says that AI systems "require one to constantly understand why the model generated the result that it produced"; a scorecard is how a forecaster does that at 3 am.
- **Probabilities first.** The one product that carried the uncertainty honestly all week was the NHC's wind-speed probability table: 56–76% is a warning and not a promise, and it fell to 8% nine hours before the closest pass. It was also the product residents were least likely to notice; our own graphics carried it as a line under the building numbers. The deterministic line and the cone should be shown as what they are, one scenario inside a distribution.

10.4 The observing network behind the forecast

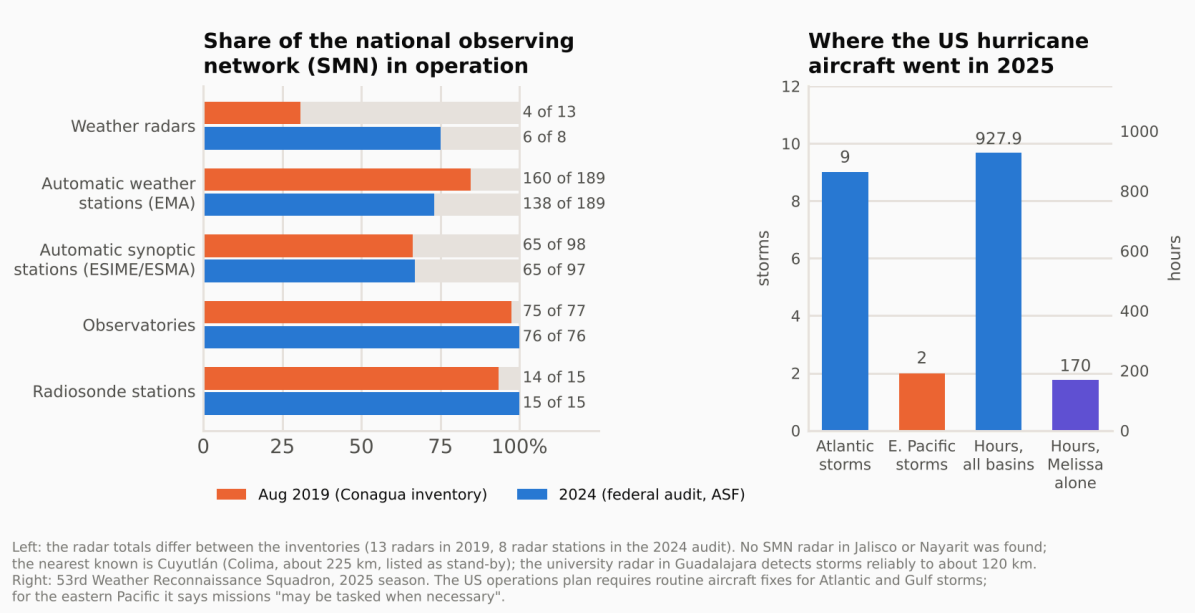


Figure 11. Left: the share of Mexico's national observing network in operation, from Conagua's 2019 inventory and the federal auditor's 2024 report. Right: the 2025 flying season of the US Air Force Reserve hurricane squadron, which covers the eastern Pacific only when tasked.

The federal auditor's report on the 2024 budget year records 85 of 387 observing systems out of service for lack of maintenance or storm damage, two of the eight weather radars down, 51 of 189 automatic stations down, and no new observatory or automatic station installed during the year. Conagua itself acknowledged in 2020 that the radar network had passed its useful life and that a 25-million-peso mobile X-band radar had been left out of the budget. No SMN radar in Jalisco or Nayarit could be found (the SMN's own radar page could not be checked); the nearest SMN radar known is Cuyutlán in Colima, about 225 km to the south-south-east, a 1994 unit the WMO radar database lists as on stand-by, and the next sites are far to the north (El Palmito in the Durango sierra, about 570 km; Guasave in Sinaloa). The Universidad de Guadalajara's own radar, about 200 km away across the Sierra, is described by the university as detecting storms reliably out to about 120 km, which does not reach the bay. Over the water the picture is thinner still. The US operations plan requires routine aircraft fixes for Atlantic and Gulf storms and says only that eastern Pacific missions "may be tasked when necessary"; for Lidia in 2023 that meant two flights and four center fixes, the last five hours before a category-4 landfall about 65 km from this city, and no coastal wind observation at all near the landfall point. In 2025 the squadron flew two eastern Pacific storms against nine in the Atlantic. This is not a criticism of the NHC, which forecasts for 28 countries with aircraft that belong to another agency, the US Air Force Reserve; it is the reason intensity at this coast rests on satellite estimates.

What is encouraging is that the ocean side has started to move. The Apixqui programme (CICESE, UNAM, the Navy and the SMN, funded by the science ministry since 2025) is deploying gliders and buoys in the Pacific, ran four glider missions off Guerrero last season, sampled Priscilla, and plans a radiosonde station; its stated motivation is Otis. A partnership on this coast should attach itself to that programme rather than duplicate it: Banderas Bay and the shelf off Cabo Corrientes are an obvious place for its buoys.

10.5 People and training

Two kinds of training are needed and they are usually confused. The first is the training of models, which needs data: the gridded AI models are trained on reanalysis at 28 km (DeepMind's cyclone model also on a database of about 5,000 historical storms), and the gridded models' known weaknesses (smoothed cores, collapsed

intensity) are weaknesses of that training set. The local layer proposed above needs its own training set, which means two or three seasons of measurements in the bay before it is worth much. That is an argument for installing the instruments first and the models second, and for starting now.

The second is the training of people. The NHC runs a two-week workshop in English and Spanish for the region's government meteorologists every year; the WMO and the SMN agreed in 2020 to work on the SMN's technical staff development; neither is specific to AI guidance or to building-scale products. What the week showed is that a forecaster, a civil protection officer and a hotel manager each need to be able to read a probability, an ensemble spread and a verification table, and to say in plain words which model to believe tonight and why. That is a short course, not a degree, and it should be taught jointly to the SMN, the state unit (UEPCBJ), the municipal civil protection unit and the hotel association's emergency coordinators before each season, with the previous season's storms as the material. The partnership should also fund three full-time posts at a local university group (the Universidad de Guadalajara's meteorology institute, or CICESE through Apixqui) whose job during a storm is to run the verification, the local model and the resident channel, and whose job the rest of the year is to make the products better. The most useful observation of this storm came from a resident who said it was not raining; a WhatsApp group of thirty people with a view of the bay, a trained moderator and a rule that every report carries a time and a place is a legitimate instrument and costs nothing.

10.6 A realistic public-private partnership

The principle has to be stated first, because it is where such partnerships go wrong: **the forecast and the warning stay public.** The SMN is the official source, the national civil protection coordination issues the SIAT-CT alert, the state and municipal units act on it, and nothing a private partner produces replaces any of that. The WMO's 2019 Geneva Declaration says the same: the national services remain the core of observation and analysis, and the private sector's growing role should be organised around them. What a partnership adds is what the public side does not have the budget or the mandate for: instruments on private property, computing, product design, distribution to visitors, and money that does not depend on the next federal budget. Figure 12 shows the shape.

Who brings what, and what they get.

Partner	Brings	Gets
SMN / Conagua	Official forecast and data; station maintenance; the scorecard's home page; a forecaster on the storm roster.	A radar and a mesonet it does not have to buy; a tested local layer; better verification of its own products.
CNPC, CENAPRED, UEPCBJ, municipal unit	Ownership of the alert and the local products; training time; the SIAT-CT link.	Building-scale guidance to act on; a record of what each alert led to.
NHC / NOAA (through WMO RA-IV)	Model access, the annual workshop, coordination on watches; possibly tasked aircraft when a storm threatens the bay.	Coastal observations it currently lacks for its own verification.
University group (UdeG, CICESE / Apixqui, UNAM)	Operations, modelling, verification, the archive; students.	Funded posts; a data set that does not exist anywhere else.
AI and cloud partner	Compute, model access, engineering help; open weights where they exist (DeepMind's cyclone model and NVIDIA's Earth-2 family are published).	A real-world test bed with ground truth; the credit for it.

Hotels, airport (GAP), port, marina, utilities	Roofs and piers for instruments; distribution to guests; protocols; part of the money.	Hours of warning on a probability they can act on; closures and reopenings timed to that probability rather than to a line (the airport closed on Saturday afternoon on a forecast of a direct hit and stayed closed through a night of 4-kt winds).
Insurers and reinsurers; the sovereign cat-bond programme	Part of the money; loss and claims data in aggregate.	Local observations for parametric triggers and pricing; fewer disputes about what the wind actually was.
Telecoms	Cell-broadcast and SMS delivery; connectivity for stations.	A defined role in the protocol.

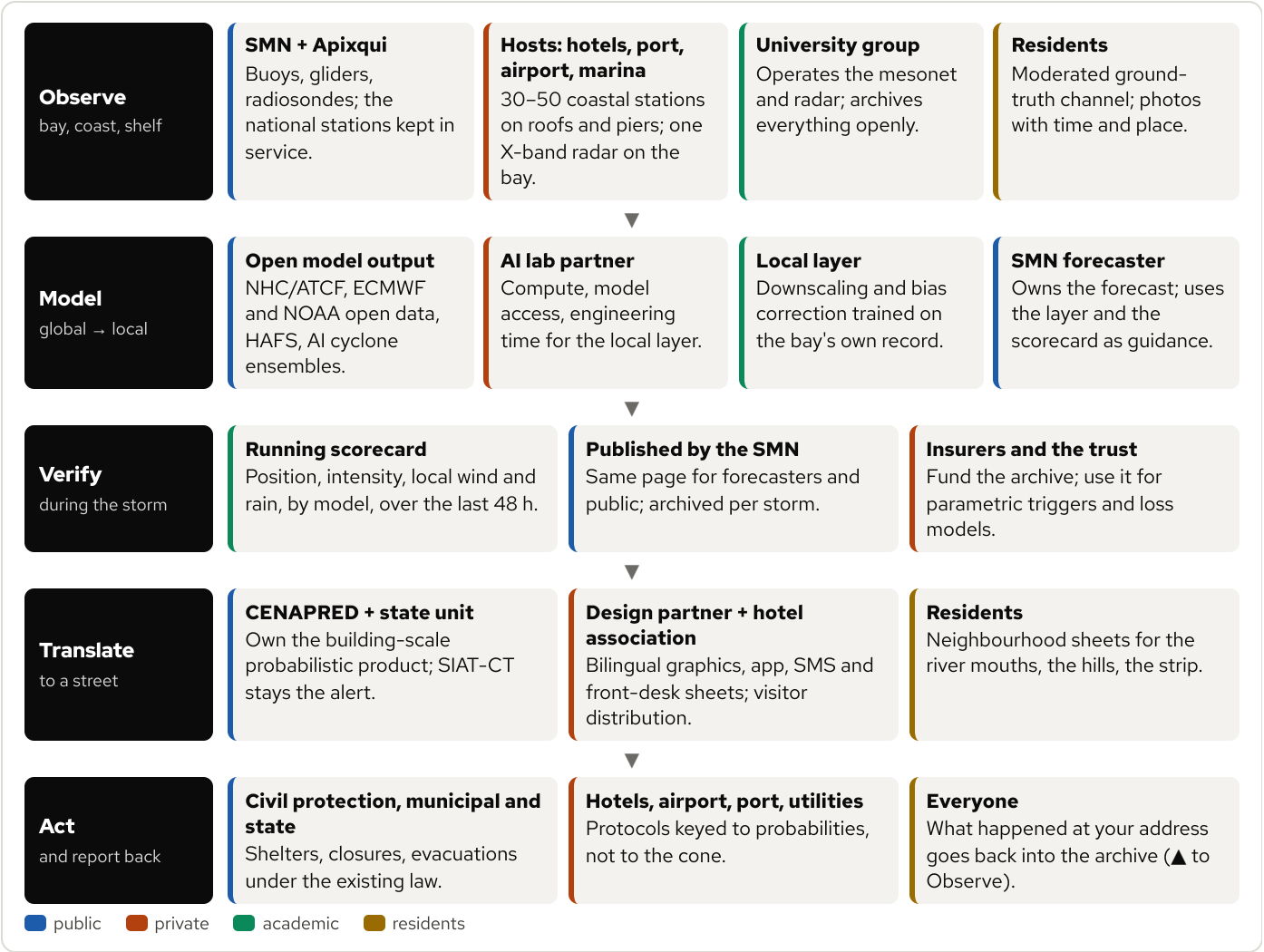


Figure 12. The shape of the partnership. Public bodies keep the forecast, the alert and the response; private partners add instruments, computing, design, distribution and money; a university group runs the technical work and the archive; residents supply ground truth and get street-level products back.

Money, and where it realistically comes from. The order of magnitude matters more than the precision, so here is mine, to be costed properly: one dual-polarisation X-band radar sited on the bay, about 25–30 million pesos installed (Conagua's own 2020 figure for a mobile unit was 25 million); 30–50 coastal stations at roughly 150,000–250,000 pesos each including telemetry and three years of maintenance; two or three buoys through Apixqui; computing in the low hundreds of thousands of pesos a year; a technical team of six to eight people (the

three university posts, a forecaster seconded from the SMN during the season, a designer-translator, and field staff for the stations); and the training programme. Over three seasons that is in the range of 70–120 million pesos (about US$4–7 million), comparable to what the SMN's budget programme for meteorological and hydrological systems spent in all of 2024 (97.5 million pesos) and a fifth to a third of one percent of the insured loss from Otis alone (35.5 billion pesos). The sources that could carry it already exist:

- **The lodging tax and a coastal trust.** Quintana Roo pays for its reef-and-beach parametric insurance through a trust fed by hotel-tax revenue, created by the state, the hotel industry and The Nature Conservancy; it paid out after Hurricane Delta. A Banderas Bay trust on the same model, drawing a small share of the state lodging taxes, would fund the operating budget without a new line in the federal budget.
- **The hotel and tourism sector directly.** The hotel association had more than 12,000 visitors in the city during Simon at about 50% occupancy; all of them were sheltered in place or offered rebooking on the strength of a forecast line that was 50–100 km off. The sector's interest in a better one is not charitable.
- **Insurers and the sovereign programme.** Mexico holds about US$1.17 billion of parametric protection, including US$595 million of World Bank catastrophe bonds, one of them specific to Pacific hurricanes; Simon's lowest pressure crossed that bond's pressure threshold while its track points stayed just outside the trigger zones, "a very close call" in the words of one analyst. Parametric products are priced on observations, and the industry has a direct interest in a coast that can measure what hits it. The NOAA model of buying private observations under contract (US$6.4 million in two radio-occultation awards announced in August 2026) is the template for the public side paying for data rather than hardware.
- **Technology partners in kind.** The Met Office–Microsoft supercomputer and the NHC–DeepMind arrangement show the two forms this takes: paid infrastructure, and model access with engineering time. Both are realistic at this scale; neither should be allowed to own the data or the product.
- **Development finance for the public core.** The World Bank financed the SMN's last modernisation (US$105 million, closed); the IDB and WMO's SOFF fund observations in the region, though SOFF prioritises least-developed and small-island states and Mexico is not among its beneficiaries. A state-level loan or an IDB technical cooperation for the Jalisco–Nayarit coast is plausible; a national programme is not what this proposal needs to wait for.

Governance. A fideicomiso with a small board (state civil protection, the SMN, the municipality, the hotel association, the university, one insurer, one technology partner), an open-data rule for every observation and every forecast product, a published verification after every storm, and an annual review with the NHC through the regional hurricane committee. The private partners get credit, data access and a seat; they do not get to issue a warning, and the agreement should say so in its first article.

10.7 Three seasons

	Season 1 (2027): measure and show	Season 2 (2028): model and train	Season 3 (2029): hand over and extend
Observations	20–30 coastal stations on hotel, port and airport roofs; two Apixqui buoys off Cabo Corrientes; the resident channel.	X-band radar on the bay; stations to 50; radiosonde through Apixqui.	Network extended south to Barra de Navidad and north to San Blas; maintenance contract transferred to the SMN.
Models	Open model feeds and the scorecard running live; parametric wind field as the interim local layer.	Local downscaling and bias correction trained on season 1; HAFS and AI ensembles blended by verified skill.	Rain and surge layers; model retrained on two seasons; code and weights published.
Products	Bilingual building-scale sheets with probabilities, issued with each advisory; hotel front-desk version.	Products adopted by CENAPRED and the state unit as official; cell broadcast and app.	Neighbourhood sheets for every colonia; regional template for other coastal cities.
People	Pre-season course for SMN, civil protection and hotel coordinators; three funded university posts; storm roster.	Second cohort; exchange week with the NHC; students on the verification.	Course run by the SMN and the state; the posts become permanent.
Money	Hotel sector and technology partners in kind; trust set up.	Trust funded from the lodging tax; insurer contribution; radar financed.	Operating budget inside the trust; capital replacement planned.
Test of success	Every advisory verified within an hour; at least one storm's local wind and rain measured end to end.	Local layer beats the raw global models on wind at the stations; forecasters use the scorecard during a storm.	A warning for the bay carries a probability; the SMN owns the system; a second city has copied it.

Figure 13. A three-season roadmap. The order is deliberate: instruments before models, because the models need the data; products from the first season, because the products are what residents and hotels will pay for.

10.8 What would make it fail

- **Treating it as a technology project.** The radar and the models are the easy part. Maintenance budgets, a duty roster, and an agreement about who may say what during a storm are what decide whether it works in year three.
- **A private forecast competing with the official one.** Two voices during a hurricane is worse than one bad one. Everything the partnership produces must be issued under the public authority or clearly labelled as guidance to it.
- **Closed data.** If the observations or the products are proprietary, the SMN cannot use them, the NHC cannot verify against them and the next storm's lessons are lost. Open by default, with the trust as custodian.
- **Over-trusting the AI models after a good year.** DeepMind's model won this storm and the 2025 season on track; its intensity skill is shorter-range, its runs have been erratic in at least one documented case, and it was unavailable in real time for part of 2025. The scorecard is the protection: weight by verified skill, every storm, not by reputation.
- **Building for Vallarta only.** The same gap exists at Manzanillo, Barra de Navidad, San Blas and Mazatlán, and the storms do not choose. The products, the code and the course should be written to be copied.

10.9 The first ninety days

Convene the hotel association, the municipal and state civil protection units, the SMN's Pacific desk and the Universidad de Guadalajara's meteorology institute before the end of the year, with Simon's record (this report) as the agenda. Agree the principle in section 10.6. Ask Apixqui where its next buoys go. Price the stations and the

radar. Draft the trust. Put the first twenty stations on roofs by May 2027, and run the scorecard and the building sheets, under the SMN's name, for the first storm of the season.

Appendix A. The resident graphics, every version, on white

Everything produced for residents during the storm, in the order it was issued, rendered in English on a white background for print. Each set is dated with the advisory it was built on; the numbers are those of that moment and were superseded by the next set. The Spanish originals remain on the shared page.

A1 · Advisory 17, first set (Sunday 9:10 am)

The first set of the day. Rain scaled to the SMN regional maximum; wind from the NHC line that passed 74 km from the city.

SIMON · PUERTO VALLARTA NHC advisory 17 · Sun Oct 11, 9 am · Vallarta time

What to expect at your building

NHC for Puerto Vallarta: **56%** tropical-storm-force wind · **13%** for 58 mph or more · **1%** hurricane

The worst: Sunday 5 pm to Monday 10 am

The center would pass ~45 mi west-southwest around Mon 1 am. Gusty wind, the heaviest rain, then the rivers: stay in.

D Terrace

Amapas, 8th floor, north-facing windows

When: strong, gusty wind **Sun 11:30 am – Mon 9 am**, strongest around **Sun 4 pm**. Heaviest rain **Sun 6 pm – Mon midnight**.

Max: sustained ~**35 mph**, gusts ~**50 mph**, from the east. Rain total ~**10 in**.

Worst impact: gusts on a high floor and flying debris: interior room, away from windows. Landslides possible from **Sun 6:30 pm**: they can cut off the street.

Casa Francisca

Emiliano Zapata, Francisca Rodríguez 336, 3rd floor

When: strong, gusty wind **Sun 11:30 am – Mon 8:30 am**, strongest around **Sun 4 pm**. Heaviest rain **Sun 6 pm – Mon midnight**. **Cuale River dangerous Sun 11:30 pm – Mon 8 am**.

Max: sustained ~**30 mph**, gusts ~**45 mph**, from the east. Rain total ~**10 in**.

Worst impact: the Cuale high in the streets below (not your floor): don't go down or cross water. Swirling gusts between buildings: falling objects. Landslides possible from **Sun 6:30 pm**: they can cut off the street.

El Pitillal (Macroplaza)

Cinco de Mayo & Calle España, 2-story house

When: strong, gusty wind **Sun 1 pm – Mon 8 am**, strongest around **Sun 6:30 pm**. Heaviest rain **Sun 10 pm – Mon 4 am**. Pitillal River high from **Sun 11:30 pm**. Water on streets from **Sun 8 pm**.

Max: sustained ~**30 mph**, gusts ~**40 mph**, from the east. Rain total ~**8 in**.

Worst impact: water in the street and the river 525 ft away: move documents and medicine upstairs, cut the power before water gets in, never cross moving water.

Site figures from the NHC wind odds and the SMN rain for Jalisco (advisory 32: over 10 in). Approximate times: they can shift with each advisory.

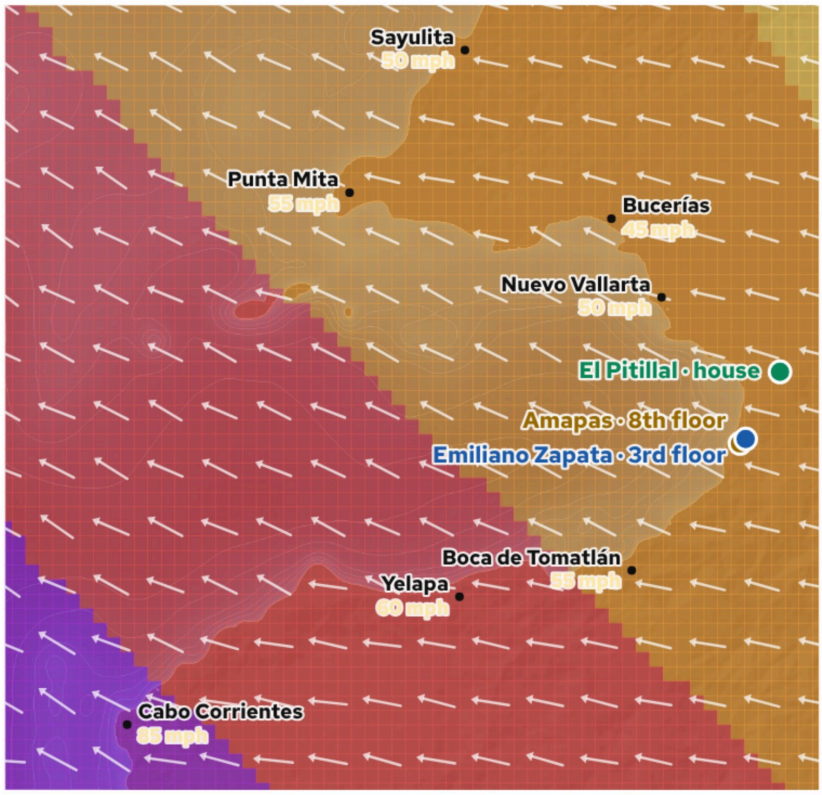
Not an official forecast · SMN · NHC · Protección Civil · 911

A1-summary. Building summary · Advisory 17, first set (Sunday 9:10 am).

SIMON · 1/6 NHC advisory 17 · Sun Oct 11, 9 am · Vallarta time

Wind: how strong, from where, where it hits hardest

Colors: peak sustained wind over open water if the center follows the NHC line exactly. Arrows: where it blows toward when strongest. White line: NHC track.



29–38 mph: strong · 39–57: tropical storm · 58–73: strong TS · 74+: hurricane

Amapas, 8th floor (Cafeto & Olas Altas)

Sustained up to **35 mph**, gusts to **50 mph**, from the **east**.
Strong wind, with tropical-storm-force gusts, **Sun 11:30 am – Mon 9 am**. Strongest around **Sun 4 pm**; the center passes around **Mon 1 am**.

Emiliano Zapata, 3rd floor (Francisca Rodríguez 336)

Sustained up to **30 mph**, gusts to **45 mph**, from the **east**.
Strong wind, with tropical-storm-force gusts, **Sun 11:30 am – Mon 8:30 am**. Strongest around **Sun 4 pm**; the center passes around **Mon 1 am**.

El Pitillal, house (Cinco de Mayo & Calle España)

Sustained up to **30 mph**, gusts to **40 mph**, from the **east**.
Strong wind, with tropical-storm-force gusts, **Sun 1 pm – Mon 8 am**. Strongest around **Sun 6:30 pm**; the center passes around **Mon 1:30 am**.

Site figures: from the NHC wind odds for Puerto Vallarta (56% tropical storm · 13% for 58 mph or more · 1% hurricane) and each building's height and surroundings. They can run higher or lower. The map does not apply that correction: it shows the open-water wind if the center followed the NHC line exactly.

Not an official forecast. Follow the SMN, the NHC and Protección Civil. Emergencies: 911.

A1-wind. Wind · Advisory 17, first set (Sunday 9:10 am).

SIMON · 2/6 NHC advisory 17 · Sun Oct 11, 9 am · Vallarta time

Rain: how much, and when it falls hardest

Total from now to Monday noon: today's satellite rain pattern carried along the NHC line, scaled to the SMN forecast for coastal Jalisco (10 in per 24 h at the wettest of the three sites). The official forecasts for Jalisco are far higher: see note.



Total over land to Monday noon: <4 in · 4–8 · 8–12 · 12–16 · >16 in

Amapas, 8th floor (Cafeto & Olas Altas)

Estimated total **10 in**. Wettest 6 hours: **Sun 6 pm – Mon midnight** (4 in).
Heavy rain **Sun 5 pm – Mon 9:30 am**; most intense around **Mon 2 am** (~0.9 in/h). The satellite alone would give ~1.7 in here; the figures follow the official forecast.

Emiliano Zapata, 3rd floor (Francisca Rodríguez 336)

Estimated total **10 in**. Wettest 6 hours: **Sun 6 pm – Mon midnight** (4 in).
Heavy rain **Sun 5 pm – Mon 9:30 am**; most intense around **Mon 2 am** (~0.9 in/h).

El Pitillal, house (Cinco de Mayo & Calle España)

Estimated total **8 in**. Wettest 6 hours: **Sun 10 pm – Mon 4 am** (3 in).
Heavy rain **Sun 6 pm – Mon 7:30 am**; most intense around **Mon 3 am** (~1.1 in/h).

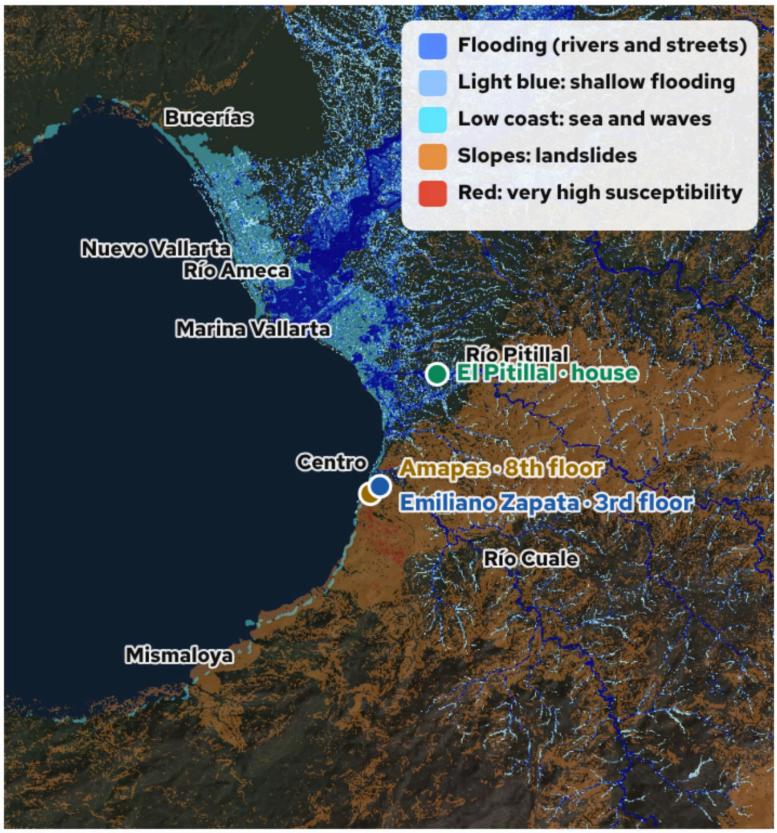
For the region: SMN (advisory 32) locally over 10 in on the Jalisco coast and west, and 3–6 in in southern Nayarit (Nuevo Vallarta, Bucerías, Punta Mita, Sayulita); NHC (advisory 17) another 8–16 in over parts of southwestern Mexico; isolated storm totals of 30 in across Michoacán, Colima and Jalisco. With more rain, the rivers rise sooner.

Not an official forecast. Follow the SMN, the NHC and Protección Civil. Emergencies: 911.

A1-rain. Rain · Advisory 17, first set (Sunday 9:10 am).

Water and slopes: where, and from when

Blue: flood-prone zones per CENAPRED (Ameca, Pitillal and Cuale rivers, canals). Light blue: low coast (under 13 ft). Orange: slopes highly susceptible to landslides.



Amapas, 8th floor (Cafeto & Olas Altas)

Water on the streets from **Sun 7 pm**. Don't cross moving water: a foot of it carries a car.
Landslides **possible from Sun 6:30 pm**, **likely from Sun 10 pm**. Keep away from cut slopes.

Emiliano Zapata, 3rd floor (Francisca Rodríguez 336)

Cuale River dangerous from Sun 11:30 pm until Mon 8 am. It rises hours after the rain on the mountains.
Landslides **possible from Sun 6:30 pm**, **likely from Sun 10:30 pm**. Keep away from cut slopes.

El Pitillal, house (Cinco de Mayo & Calle España)

Pitillal River running high from Sun 11:30 pm; dangerous if it rains more than forecast. It rises hours after the rain on the mountains.
Water on the streets from **Sun 8 pm**. Don't cross moving water: a foot of it carries a car.

Times computed with up to 10 in of rain in 24 h (10 in at Amapas and Emiliano Zapata, 8 in at El Pitillal), the SMN maximum for Jalisco (advisory 32). With more rain, everything happens sooner. Zones: Puerto Vallarta 2024 risk atlas and CENAPRED.

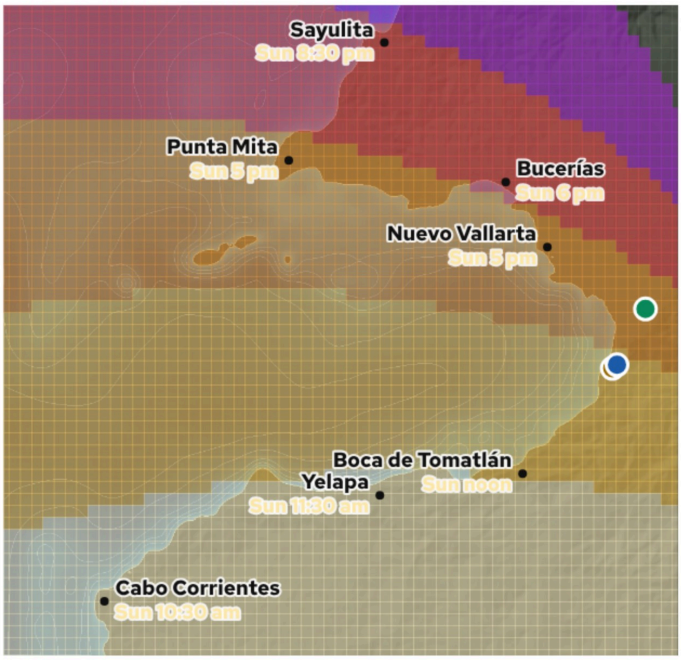
Not an official forecast. Follow the SMN, the NHC and Protección Civil. Emergencies: 911.

A1-flood. Water and slopes · Advisory 17, first set (Sunday 9:10 am).

SIMON · 4/6 NHC advisory 17 · Sun Oct 11, 9 am · Vallarta time

When it arrives, and when it is worst

Map: when tropical-storm-force wind (39 mph) reaches each part of the bay if the center follows the NHC line. Below: hour by hour at the three sites.



39-mph wind arrives

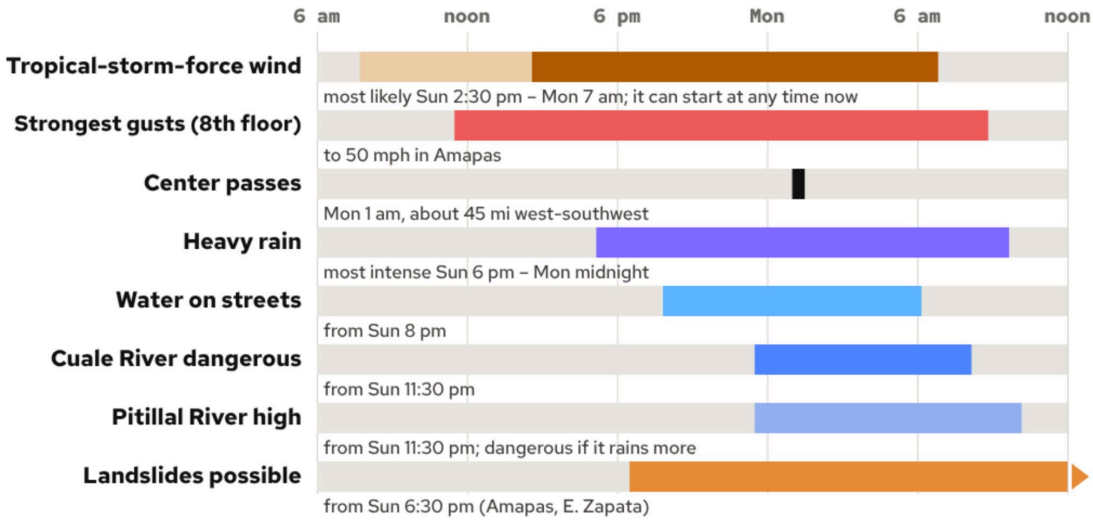
- **before Sun noon**
- **Sun noon–3 pm**
- **Sun 3–6 pm**
- **Sun 6–9 pm**
- **Sun 9 pm or later**
- **never reaches 39 mph**

Puerto Vallarta, per the NHC:

Tropical-storm-force wind: **56%** chance. Most likely from **Sun 2:30 pm**; **it can start at any time now**.

Hurricane-force wind: **1%**. The center would pass about **45 mi west-southwest**, around **Mon 1 am**.

If you still have preparations to make, **finish them now**: strong wind can start at any time.



The worst: Sunday 5 pm to Monday 10 am

Gusty wind, the heaviest rain, then the rivers. Stay in during those hours.

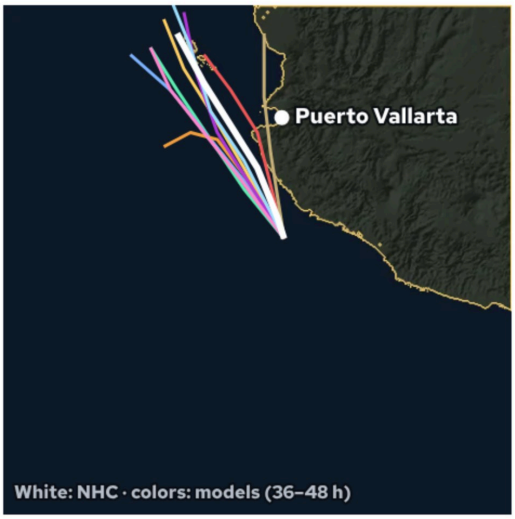
Approximate: times can shift by a few hours with each advisory. Light band: when it could start; bar: most likely. Arrow: continues past Monday noon.

Not an official forecast. Follow the SMN, the NHC and Protección Civil. Emergencies: 911.

A1-timing. Timing · Advisory 17, first set (Sunday 9:10 am).

The models: which have been right, and what they say now

Score: each model's mean position error in its Simon forecasts at 12–36 h, against what happened. Bars: peak sustained wind in downtown Vallarta if the center followed that model's line.

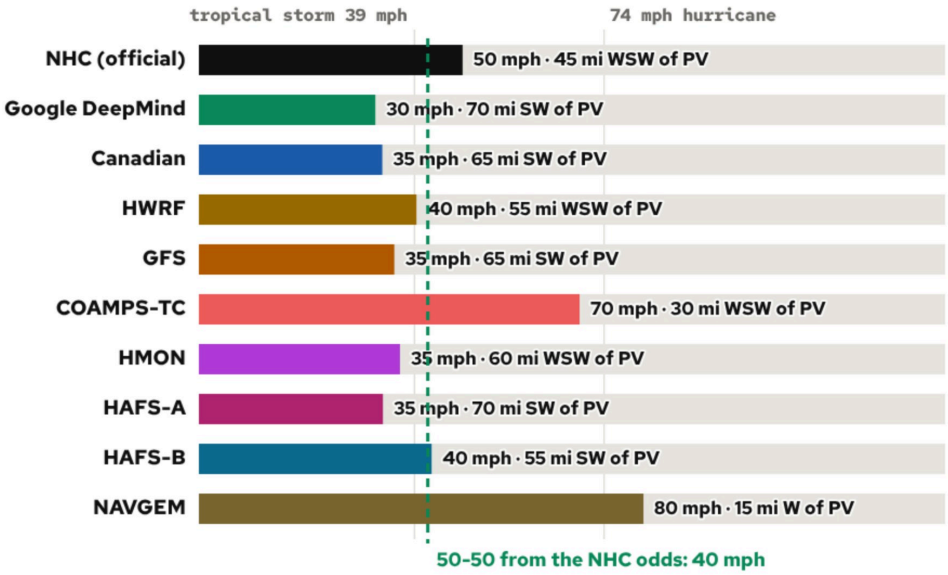


Mean position error

last 2 days · whole storm

NHC (official)	20 mi · 35 mi
Google DeepMind	25 mi · 30 mi
Canadian	35 mi · 75 mi
HWRF	40 mi · 70 mi
GFS (US)	40 mi · 70 mi
COAMPS-TC	40 mi · 65 mi
HMON	60 mi · 60 mi
HAFS-A	65 mi · 60 mi
HAFS-B	75 mi · 60 mi
NAVGEM (US Navy)	85 mi · 55 mi

Peak sustained wind in downtown Vallarta, model by model, if the center followed its line (not the NHC odds)



Reading it: the forecasts that have tracked Simon best (the NHC and Google DeepMind) keep the center 45 mi–70 mi southwest of Vallarta. None passes it over the city. The two that bring it closest, NAVGEM (15 mi) and COAMPS-TC (30 mi), are the ones giving 80 mph and 70 mph; over the last two days their mean error was 85 mi and 40 mi. No single model decides: the NHC gives Vallarta **1%** for hurricane-force wind and **56%** for tropical-storm force.

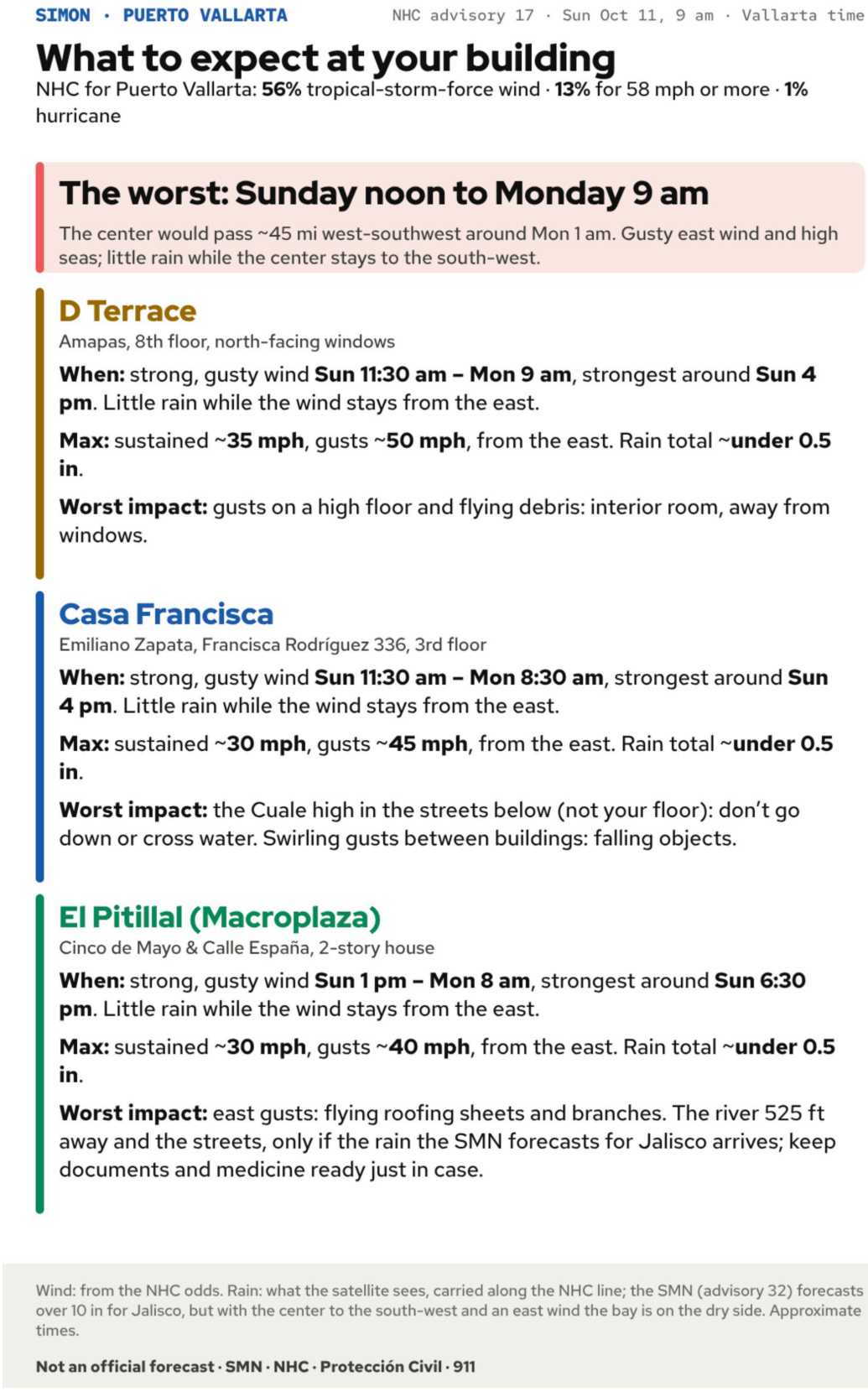
Models from the 12Z 11 Oct cycle (NHC/ATCF). Error: against the observed official track; "last 2 days" averages only 2 to 7 forecasts per model and lead time, so the order can change. Vallarta wind: the NHC wind structure on each model's line, without the odds-based correction.

Not an official forecast. Follow the SMN, the NHC and Protección Civil. Emergencies: 911.

A1-models. Models · Advisory 17, first set (Sunday 9:10 am).

A2 · Advisory 17, corrected set (Sunday 1 pm)

Same advisory, rain no longer scaled up after a ground report that it was not raining; dry-side wording added.



SIMON · PUERTO VALLARTA

NHC advisory 17 · Sun Oct 11, 9 am · Vallarta time

What to expect at your building

NHC for Puerto Vallarta: **56%** tropical-storm-force wind · **13%** for 58 mph or more · **1%** hurricane

The worst: Sunday noon to Monday 9 am

The center would pass ~45 mi west-southwest around Mon 1 am. Gusty east wind and high seas; little rain while the center stays to the south-west.

D Terrace

Amapas, 8th floor, north-facing windows

When: strong, gusty wind **Sun 11:30 am – Mon 9 am**, strongest around **Sun 4 pm**. Little rain while the wind stays from the east.

Max: sustained ~**35 mph**, gusts ~**50 mph**, from the east. Rain total ~**under 0.5 in**.

Worst impact: gusts on a high floor and flying debris: interior room, away from windows.

Casa Francisca

Emiliano Zapata, Francisca Rodríguez 336, 3rd floor

When: strong, gusty wind **Sun 11:30 am – Mon 8:30 am**, strongest around **Sun 4 pm**. Little rain while the wind stays from the east.

Max: sustained ~**30 mph**, gusts ~**45 mph**, from the east. Rain total ~**under 0.5 in**.

Worst impact: the Cuale high in the streets below (not your floor): don't go down or cross water. Swirling gusts between buildings: falling objects.

El Pitillal (Macroplaza)

Cinco de Mayo & Calle España, 2-story house

When: strong, gusty wind **Sun 1 pm – Mon 8 am**, strongest around **Sun 6:30 pm**. Little rain while the wind stays from the east.

Max: sustained ~**30 mph**, gusts ~**40 mph**, from the east. Rain total ~**under 0.5 in**.

Worst impact: east gusts: flying roofing sheets and branches. The river 525 ft away and the streets, only if the rain the SMN forecasts for Jalisco arrives; keep documents and medicine ready just in case.

Wind: from the NHC odds. Rain: what the satellite sees, carried along the NHC line; the SMN (advisory 32) forecasts over 10 in for Jalisco, but with the center to the south-west and an east wind the bay is on the dry side. Approximate times.

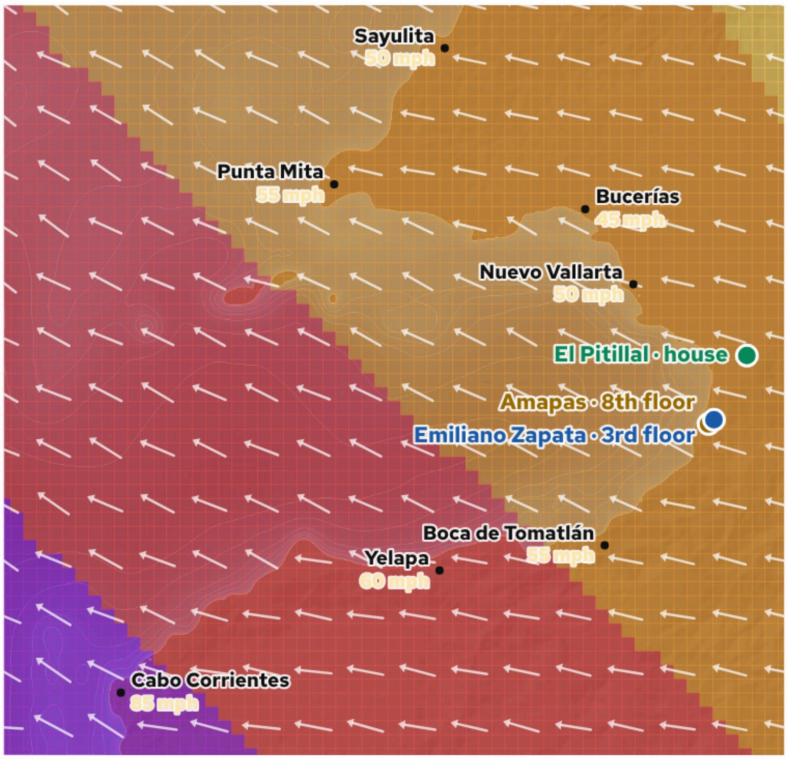
Not an official forecast · SMN · NHC · Protección Civil · 911

A2-summary. Building summary · Advisory 17, corrected set (Sunday 1 pm).

SIMON · 1/6 NHC advisory 17 · Sun Oct 11, 9 am · Vallarta time

Wind: how strong, from where, where it hits hardest

Colors: peak sustained wind over open water if the center follows the NHC line exactly. Arrows: where it blows toward when strongest. White line: NHC track.



29–38 mph: strong · 39–57: tropical storm · 58–73: strong TS · 74+: hurricane

Amapas, 8th floor (Cafeto & Olas Altas)

Sustained up to **35 mph**, gusts to **50 mph**, from the **east**.
Strong wind, with tropical-storm-force gusts, **Sun 11:30 am – Mon 9 am**. Strongest around **Sun 4 pm**; the center passes around **Mon 1 am**.

Emiliano Zapata, 3rd floor (Francisca Rodríguez 336)

Sustained up to **30 mph**, gusts to **45 mph**, from the **east**.
Strong wind, with tropical-storm-force gusts, **Sun 11:30 am – Mon 8:30 am**. Strongest around **Sun 4 pm**; the center passes around **Mon 1 am**.

El Pitillal, house (Cinco de Mayo & Calle España)

Sustained up to **30 mph**, gusts to **40 mph**, from the **east**.
Strong wind, with tropical-storm-force gusts, **Sun 1 pm – Mon 8 am**. Strongest around **Sun 6:30 pm**; the center passes around **Mon 1:30 am**.

Site figures: from the NHC wind odds for Puerto Vallarta (56% tropical storm · 13% for 58 mph or more · 1% hurricane) and each building's height and surroundings. They can run higher or lower. The map does not apply that correction: it shows the open-water wind if the center followed the NHC line exactly.

Not an official forecast. Follow the SMN, the NHC and Protección Civil. Emergencies: 911.

A2-wind. Wind · Advisory 17, corrected set (Sunday 1 pm).

SIMON · 2/6 NHC advisory 17 · Sun Oct 11, 9 am · Vallarta time

Rain: how much, and when it falls hardest

Total from now to Monday noon: today's satellite rain pattern carried along the NHC line (unscaled: the satellite sees no rain over the bay now). The official forecasts for Jalisco are far higher: see note.



Total over land to Monday noon: <4 in · 4–8 · 8–12 · 12–16 · >16 in

Amapas, 8th floor (Cafeto & Olas Altas)

Estimated total: **under 0.5 in** while the wind stays from the east.

With the center to the south-west, the wind comes over the Sierra (from the east) and sinks dry into the bay: Simon's rain is over the ocean. This changes if the center comes closer or the wind turns from the south-west.

Emiliano Zapata, 3rd floor (Francisca Rodríguez 336)

Estimated total: **under 0.5 in** while the wind stays from the east.

El Pitillal, house (Cinco de Mayo & Calle España)

Estimated total: **under 0.5 in** while the wind stays from the east.

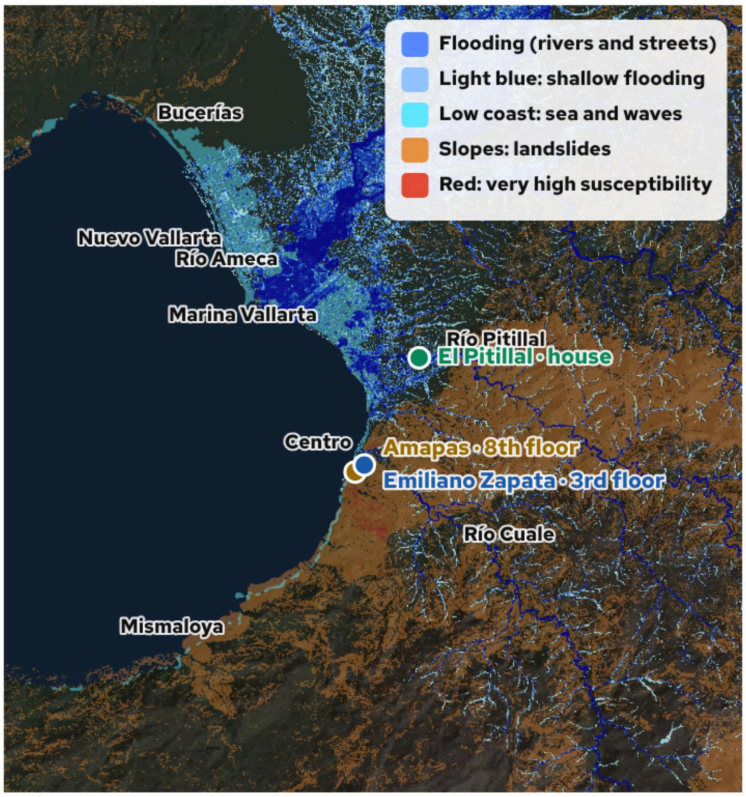
For the region: SMN (advisory 32) locally over 10 in on the Jalisco coast and west, and 3–6 in in southern Nayarit (Nuevo Vallarta, Bucerías, Punta Mita, Sayulita); NHC (advisory 17) another 8–16 in over parts of southwestern Mexico; isolated storm totals of 30 in across Michoacán, Colima and Jalisco. With more rain, the rivers rise sooner.

Not an official forecast. Follow the SMN, the NHC and Protección Civil. Emergencies: 911.

A2-rain. Rain · Advisory 17, corrected set (Sunday 1 pm).

Water and slopes: where, and from when

Blue: flood-prone zones per CENAPRED (Ameca, Pitillal and Cuale rivers, canals). Light blue: low coast (under 13 ft). Orange: slopes highly susceptible to landslides.



Amapas, 8th floor (Cafeto & Olas Altas)

With the rain the satellite sees now, **neither the river, the streets nor the slopes reach the danger threshold** in this calculation. That changes if it rains far more than the satellite sees: the SMN still forecasts extraordinary rain for Jalisco.

Emiliano Zapata, 3rd floor (Francisca Rodríguez 336)

With the rain the satellite sees now, **neither the river, the streets nor the slopes reach the danger threshold** in this calculation. That changes if it rains far more than the satellite sees: the SMN still forecasts extraordinary rain for Jalisco.

El Pitillal, house (Cinco de Mayo & Calle España)

With the rain the satellite sees now, **neither the river, the streets nor the slopes reach the danger threshold** in this calculation. That changes if it rains far more than the satellite sees: the SMN still forecasts extraordinary rain for Jalisco.

Times computed with the rain the satellite sees, carried along the NHC line (~under 0.5 in in the city). The SMN (advisory 32) still forecasts over 10 in for Jalisco: with the center to the south-west and an east wind, the bay is on the dry side; if that changes, the rivers do respond. Zones: Puerto Vallarta 2024 risk atlas and CENAPRED.

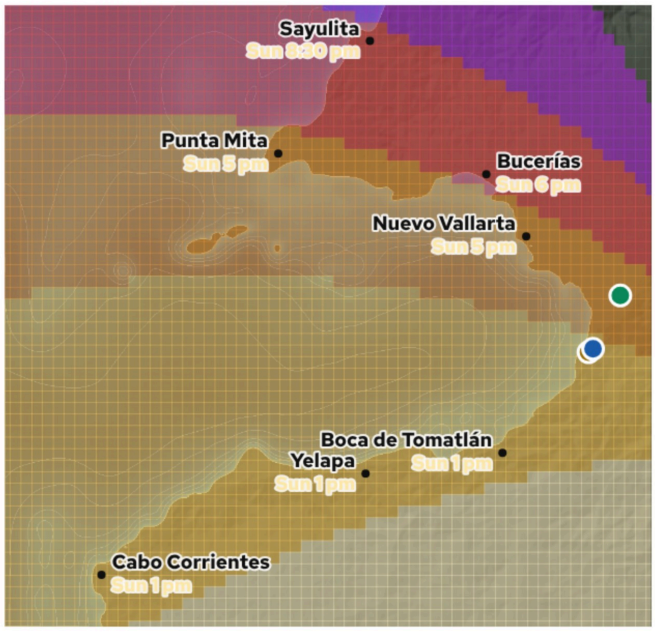
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A2-flood. Water and slopes · Advisory 17, corrected set (Sunday 1 pm).

SIMON · 4/6 NHC advisory 17 · Sun Oct 11, 9 am · Vallarta time

When it arrives, and when it is worst

Map: when tropical-storm-force wind (39 mph) reaches each part of the bay if the center follows the NHC line. Below: hour by hour at the three sites.



39-mph wind arrives

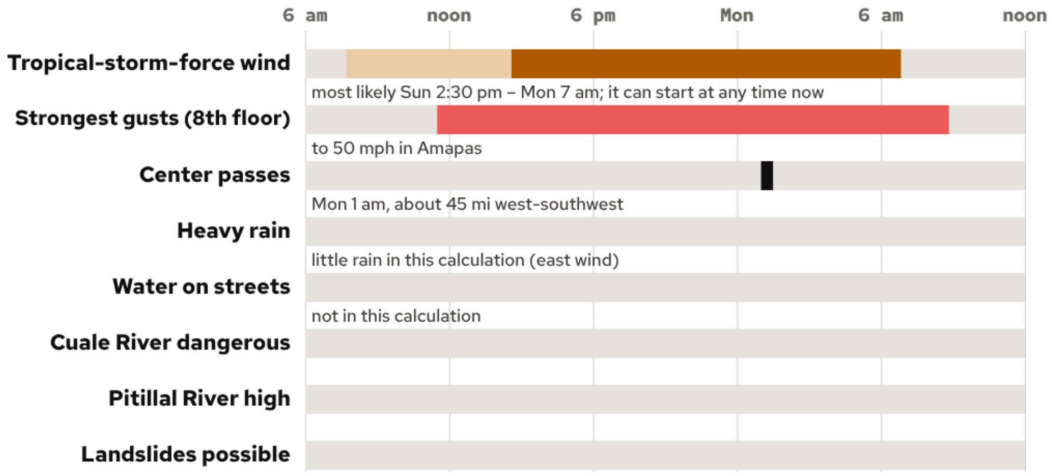
- **before Sun noon**
- **Sun noon–3 pm**
- **Sun 3–6 pm**
- **Sun 6–9 pm**
- **Sun 9 pm or later**
- **never reaches 39 mph**

Puerto Vallarta, per the NHC:

Tropical-storm-force wind: **56%** chance. Most likely from **Sun 2:30 pm**; **it can start at any time now**.

Hurricane-force wind: **1%**. The center would pass about **45 mi west-southwest**, around **Mon 1 am**.

If you still have preparations to make, **finish them now**: strong wind can start at any time.



The worst: Sunday noon to Monday 9 am

East gusts and high seas; little rain while the center stays to the south-west.

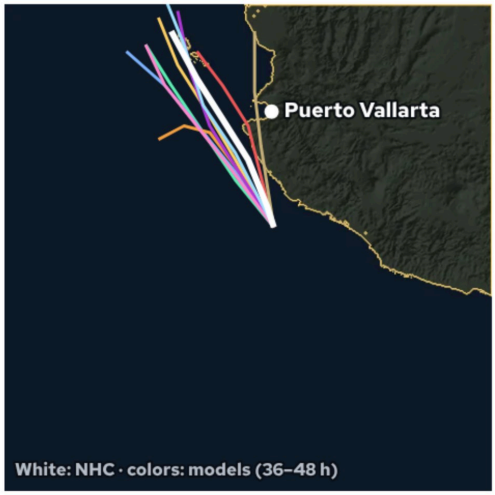
Approximate: times can shift by a few hours with each advisory. Light band: when it could start; bar: most likely. Arrow: continues past Monday noon.

Not an official forecast. Follow the SMN, the NHC and Protección Civil. Emergencies: 911.

A2-timing. Timing · Advisory 17, corrected set (Sunday 1 pm).

The models: which have been right, and what they say now

Score: each model's mean position error in its Simon forecasts at 12–36 h, against what happened. Bars: peak sustained wind in downtown Vallarta if the center followed that model's line.

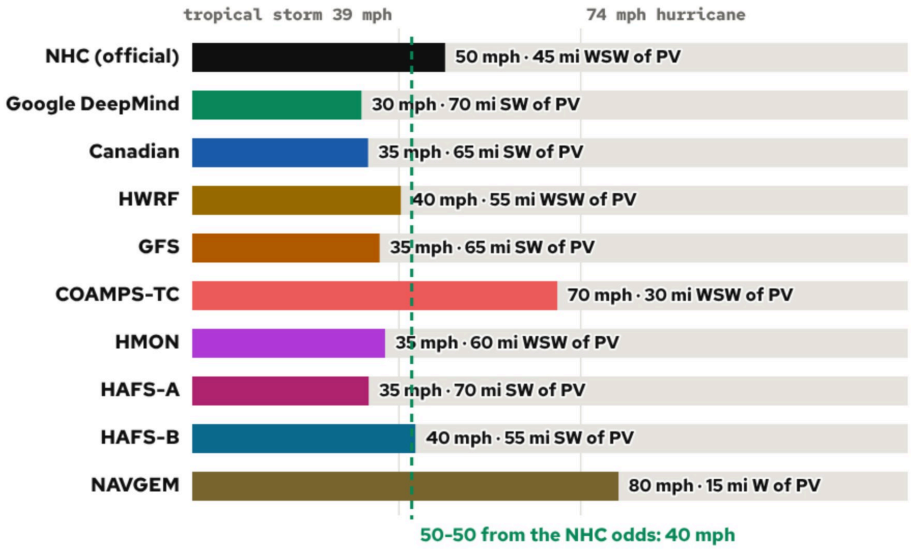


Mean position error

last 2 days · whole storm

NHC (official)	20 mi · 35 mi
Google DeepMind	25 mi · 30 mi
Canadian	35 mi · 75 mi
HWRF	40 mi · 70 mi
GFS (US)	40 mi · 70 mi
COAMPS-TC	40 mi · 65 mi
HMON	60 mi · 60 mi
HAFS-A	65 mi · 60 mi
HAFS-B	75 mi · 60 mi
NAVGEM (US Navy)	85 mi · 55 mi

Peak sustained wind in downtown Vallarta, model by model, if the center followed its line (not the NHC odds)



Reading it: the forecasts that have tracked Simon best (the NHC and Google DeepMind) keep the center 45 mi–70 mi southwest of Vallarta. None passes it over the city. The two that bring it closest, NAVGEM (15 mi) and COAMPS-TC (30 mi), are the ones giving 80 mph and 70 mph; over the last two days their mean error was 85 mi and 40 mi. No single model decides: the NHC gives Vallarta **1%** for hurricane-force wind and **56%** for tropical-storm force.

Models from the 12Z 11 Oct cycle (NHC/ATCF). Error: against the observed official track; "last 2 days" averages only 2 to 7 forecasts per model and lead time, so the order can change. Vallarta wind: the NHC wind structure on each model's line, without the odds-based correction.

Not an official forecast. Follow the SMN, the NHC and Protección Civil. Emergencies: 911.

A2-models. Models · Advisory 17, corrected set (Sunday 1 pm).

A3 · Advisory 18 (Sunday 3 pm)

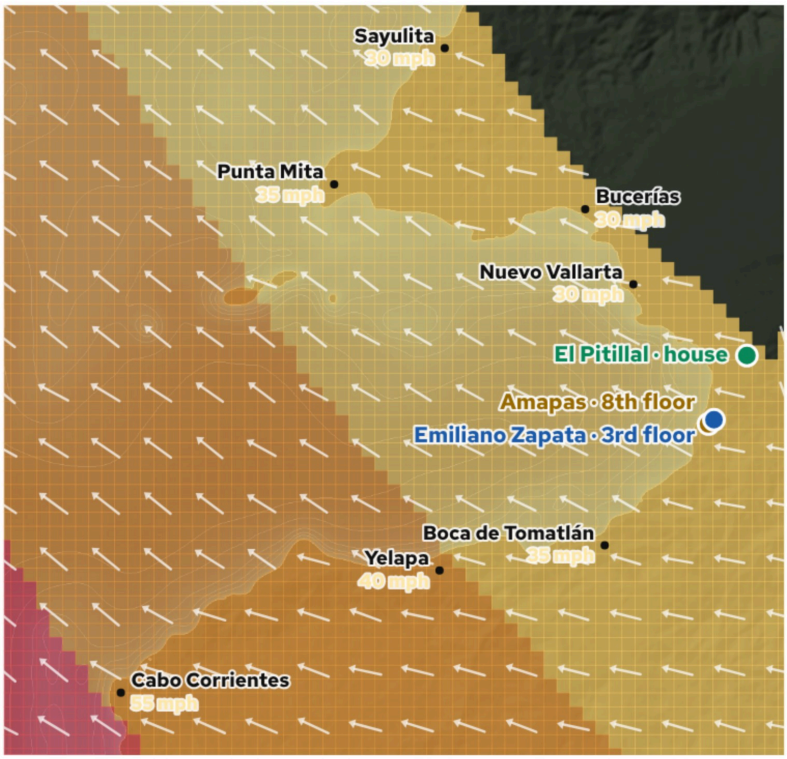
The last set before the closest pass: the NHC line at 101 km, Vallarta tropical-storm odds 8%. Includes the model verification graphic.



A3-summary. Building summary · Advisory 18 (Sunday 3 pm).

Wind: how strong, from where, where it hits hardest

Colors: peak sustained wind over open water if the center follows the NHC line exactly. Arrows: where it blows toward when strongest. White line: NHC track.



29–38 mph: strong · 39–57: tropical storm · 58–73: strong TS · 74+: hurricane

Amapas, 8th floor (Cafeto & Olas Altas)

Sustained up to **25 mph**, gusts to **40 mph**, from the **east**.
Strong, gusty wind **Sun 3 pm – Mon 4 am**. Strongest around **Sun 7:30 pm**; the center passes around **Mon midnight**.

Emiliano Zapata, 3rd floor (Francisca Rodríguez 336)

Sustained up to **25 mph**, gusts to **35 mph**, from the **east**.
Strong, gusty wind **Sun 3 pm – Mon 4:30 am**. Strongest around **Sun 7 pm**; the center passes around **Mon midnight**.

El Pitillal, house (Cinco de Mayo & Calle España)

Sustained up to **20 mph**, gusts to **30 mph**, from the **east**.
Strong, gusty wind **Sun 3 pm – Mon 4:30 am**. Strongest around **Sun 8 pm**; the center passes around **Mon midnight**.

Site figures: from the NHC wind odds for Puerto Vallarta (8% tropical storm · under 3% for 58 mph or more · under 1% hurricane) and each building's height and surroundings. They can run higher or lower. The map does not apply that correction: it shows the open-water wind if the center followed the NHC line exactly.

Not an official forecast. Follow the SMN, the NHC and Protección Civil. Emergencies: 911.

A3-wind. Wind · Advisory 18 (Sunday 3 pm).

SIMON · 2/6 NHC advisory 18 · Sun Oct 11, 3 pm · Vallarta time

Rain: how much, and when it falls hardest

Total from now to Monday noon: today's satellite rain pattern carried along the NHC line (unscaled: the satellite sees no rain over the bay now). The official forecasts for Jalisco are far higher: see note.



Total over land to Monday noon: <4 in · 4–8 · 8–12 · 12–16 · >16 in

Amapas, 8th floor (Cafeto & Olas Altas)

Estimated total: **under 0.5 in** while the wind stays from the east.

With the center to the south-west, the wind comes over the Sierra (from the east) and sinks dry into the bay: Simon's rain is over the ocean. This changes if the center comes closer or the wind turns from the south-west.

Emiliano Zapata, 3rd floor (Francisca Rodríguez 336)

Estimated total: **under 0.5 in** while the wind stays from the east.

El Pitillal, house (Cinco de Mayo & Calle España)

Estimated total: **under 0.5 in** while the wind stays from the east.

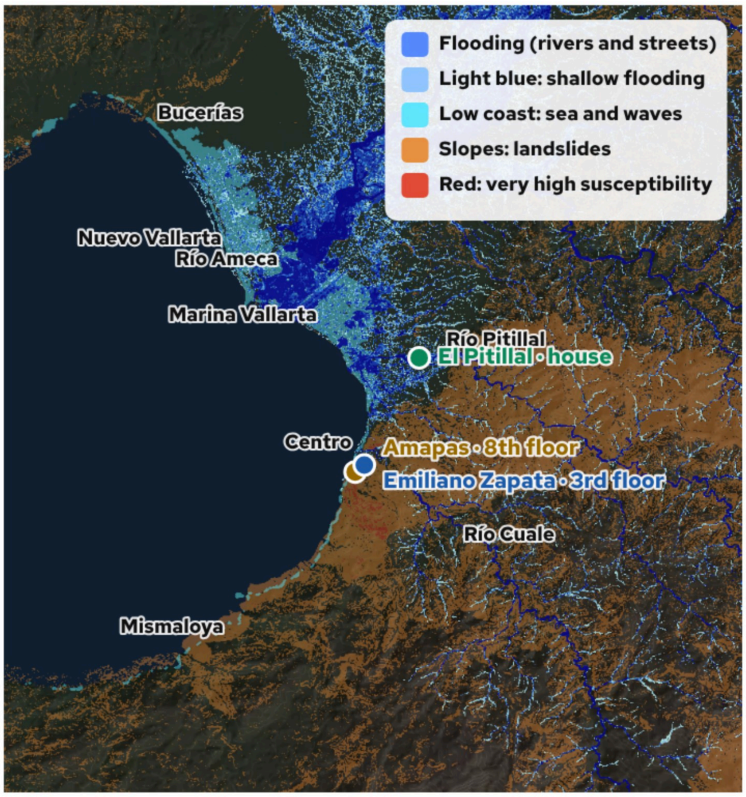
For the region: SMN (advisory 34) locally over 10 in on the Jalisco coast and west, and 3–6 in in southern Nayarit (Nuevo Vallarta, Bucerías, Punta Mita, Sayulita); NHC (advisory 18) another 2–4 in over parts of southwestern Mexico. With more rain, the rivers rise sooner.

Not an official forecast. Follow the SMN, the NHC and Protección Civil. Emergencies: 911.

A3-rain. Rain · Advisory 18 (Sunday 3 pm).

Water and slopes: where, and from when

Blue: flood-prone zones per CENAPRED (Ameca, Pitillal and Cuale rivers, canals). Light blue: low coast (under 13 ft). Orange: slopes highly susceptible to landslides.



Amapas, 8th floor (Cafeto & Olas Altas)

With the rain the satellite sees now, **neither the river, the streets nor the slopes reach the danger threshold** in this calculation. That changes if it rains far more than the satellite sees: the SMN still forecasts extraordinary rain for Jalisco.

Emiliano Zapata, 3rd floor (Francisca Rodríguez 336)

With the rain the satellite sees now, **neither the river, the streets nor the slopes reach the danger threshold** in this calculation. That changes if it rains far more than the satellite sees: the SMN still forecasts extraordinary rain for Jalisco.

El Pitillal, house (Cinco de Mayo & Calle España)

With the rain the satellite sees now, **neither the river, the streets nor the slopes reach the danger threshold** in this calculation. That changes if it rains far more than the satellite sees: the SMN still forecasts extraordinary rain for Jalisco.

Times computed with the rain the satellite sees, carried along the NHC line (~under 0.5 in in the city). The SMN (advisory 34) still forecasts over 10 in for Jalisco: with the center to the south-west and an east wind, the bay is on the dry side; if that changes, the rivers do respond. Zones: Puerto Vallarta 2024 risk atlas and CENAPRED.

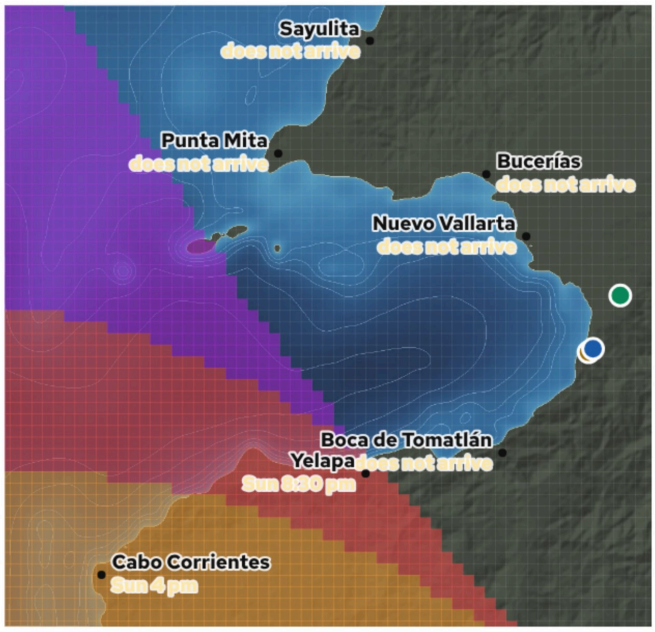
Not an official forecast. Follow the SMN, the NHC and Protección Civil. Emergencies: 911.

A3-flood. Water and slopes · Advisory 18 (Sunday 3 pm).

SIMON · 4/6 NHC advisory 18 · Sun Oct 11, 3 pm · Vallarta time

When it arrives, and when it is worst

Map: when tropical-storm-force wind (39 mph) reaches each part of the bay if the center follows the NHC line. Below: hour by hour at the three sites.



39-mph wind arrives

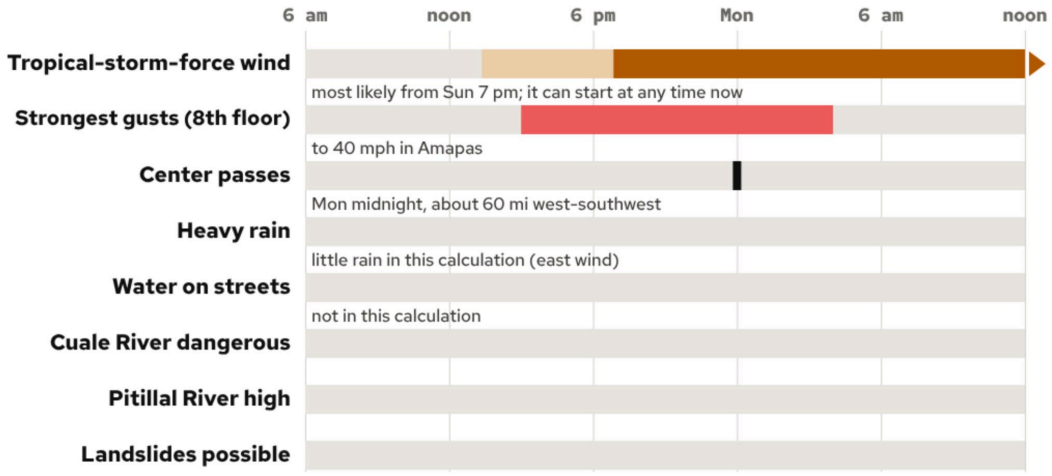
- before Sun noon
- Sun noon–3 pm
- Sun 3–6 pm
- Sun 6–9 pm
- Sun 9 pm or later
- never reaches 39 mph

Puerto Vallarta, per the NHC:

Tropical-storm-force wind: **8%** chance. Most likely from **Sun 7 pm**; **it can start at any time now**.

Hurricane-force wind: **under 1%**. The center would pass about **60 mi west-southwest**, around **Mon midnight**.

If you still have preparations to make, **finish them now**: strong wind can start at any time.



The worst: Sunday 3 pm to Monday 4 am

East gusts and high seas; little rain while the center stays to the south-west.

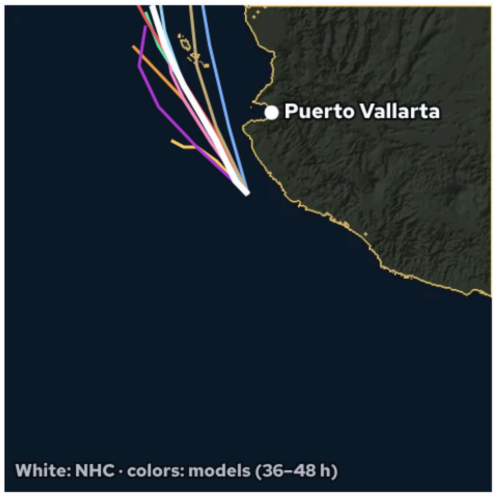
Approximate: times can shift by a few hours with each advisory. Light band: when it could start; bar: most likely. Arrow: continues past Monday noon.

Not an official forecast. Follow the SMN, the NHC and Protección Civil. Emergencies: 911.

A3-timing. Timing · Advisory 18 (Sunday 3 pm).

The models: which have been right, and what they say now

Score: each model's mean position error in its Simon forecasts at 12–36 h, against what happened. Bars: peak sustained wind in downtown Vallarta if the center followed that model's line.

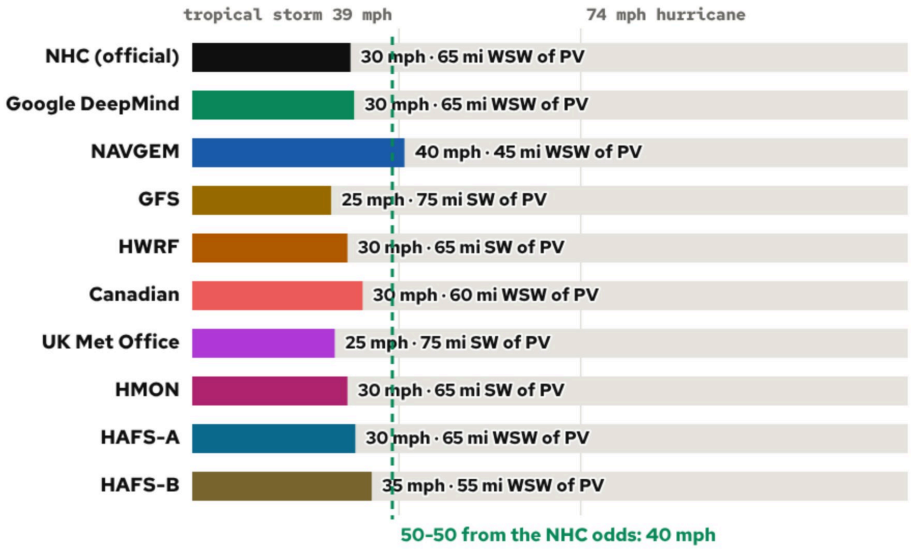


Mean position error

last 2 days · whole storm

Google DeepMind	20 mi · 30 mi
NHC (official)	30 mi · 35 mi
NAVGEM (US Navy)	40 mi‡ · 55 mi
GFS (US)	40 mi · 65 mi
HWRF	40 mi · 70 mi
Canadian	45 mi · 75 mi
UK Met Office	50 mi · 85 mi
HMON	60 mi · 60 mi
HAFS-A	75 mi · 60 mi
HAFS-B	105 mi · 60 mi

Peak sustained wind in downtown Vallarta, model by model, if the center followed its line (not the NHC odds)



Reading it: the forecasts that have tracked Simon best (Google DeepMind and the NHC) keep the center about 65 mi southwest of Vallarta. None passes it over the city. The two that bring it closest, NAVGEM (45 mi) and HAFS-B (55 mi), are the ones giving 40 mph and 35 mph; over the last two days their mean error was 40 mi‡ and 105 mi. No single model decides: the NHC gives Vallarta **under 1%** for hurricane-force wind and **8%** for tropical-storm force.

Models from the 18Z 11 Oct cycle (NHC/ATCF). Error: against the observed official track; "last 2 days" averages only 1 to 7 forecasts per model and lead time, so the order can change. Vallarta wind: the NHC wind structure on each model's line, without the odds-based correction.

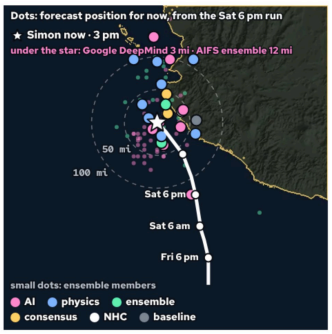
Not an official forecast. Follow the SMN, the NHC and Protección Civil. Emergencies: 911.

A3-models. Models · Advisory 18 (Sunday 3 pm).

SIMON · 6/6 NHC advisory 18 · Sun Oct 11, 3 pm · Vallarta time

Which model got it right? The last 2 days, checked

Each model against what happened: where it put Simon (versus the NHC's official position) and the wind and rain it gave Vallarta, versus the airport record. Reference run: Sat 6 pm, checked today at 3 pm.



Position · mean error at 12–36 h, last 2 days
1. Google DeepMind (21 mi) **2.** Corrected consensus (24 mi) **3.** NHC (29 mi). Best physics model (fully verified): GFS (42 mi); best ensemble: ECMWF ensemble mean (34 mi).

Right now (3 pm) · what the Sat 6 pm run said
Closest: Google DeepMind 3 mi · ECMWF AIFS ensemble 12 mi · ECMWF IFS 15 mi. **Farthest:** AIGFS 131 mi N · FourCastNet 122 mi SSE · UKMET 111 mi NNE. Ensemble members within 62 mi (100 km) of the real center: GEFS 11 of 20; ECMWF AIFS ensemble 43 of 51.

Vallarta wind · forecast versus measured
Measured: max wind 13 kt (15 mph) at the airport (METAR, 48 h). Closest gridded model: **HAFS-A** (16 mph); 7 gridded models (GFS, GraphCast, FourCastNet…) gave 5–8 mph (too low: the bay does not fit a 15-mile grid). With its own wind structure, DeepMind implied 29 mph and over open water off the city; track-based models as a group implied 25–46 mph: too high.

Vallarta rain · forecast versus measured
Measured: no rain until 1 pm, then one shower (~0.35 in at the airport). Forecast for the same period: AIGFS 0.4 in · GFS 0.7 in · GEFS ensemble mean 0.7 in · HAFS-A 1.0 in · GraphCast 1.5 in. AIGFS got it right; the rest were too wet, GraphCast by a lot. DeepMind's cyclone model gives no rain (position, intensity and wind radii only); its gridded model is GraphCast.

Every model, one by one

	Mean error 12–36 h · 2 days	Where it put it today Sat 6 pm run	Max PV wind measured 15 mph	PV rain measured 0–0.4 in
Google DeepMind · AI	21 mi	3 mi E	29 mph*	–
Corrected consensus (NHC)	24 mi	21 mi NE	38 mph*	–
NHC (official forecast)	29 mi	41 mi ENE	44 mph*	–
ECMWF AIFS · AI	31 mi	35 mi SSE	–	–
Track consensus	32 mi	45 mi NNE	–	–
Climatology + persistence	33 mi	60 mi E	–	–
ECMWF ensemble mean	34 mi	32 mi SSE	28 mph*	–
GEFS ensemble mean (US)	35 mi	30 mi NNE	6 mph	0.7 in
ECMWF AIFS ensemble · AI	36 mi	12 mi SW	–	–
NAVGEM (US Navy)	40 mi‡	61 mi ESE	25 mph*	–
AIGFS (NOAA) · AI	41 mi	131 mi N	5 mph	0.4 in
GraphCast (Google) · AI	41 mi‡	44 mi ENE	7 mph	1.5 in
GFS (US)	42 mi	59 mi N	8 mph	0.7 in
HWRF (US hurricane model)	42 mi	35 mi NW	39 mph*	–
GEM (Canada)	43 mi	22 mi SSE	26 mph*	–
ECMWF IFS (Europe)	43 mi	15 mi WNW	–	–
UKMET (UK)	52 mi	111 mi NNE	39 mph*	–
HMON (US hurricane model)	62 mi	–	43 mph*	–
Pangu-Weather (Huawei) · AI	67 mi‡	37 mi E	6 mph	–
HAFS-A (US hurricane model)	73 mi	103 mi NNW	16 mph	1.0 in
FourCastNet (NVIDIA) · AI	73 mi‡	122 mi SSE	7 mph	–
COAMPS-TC (US Navy)	80 mi	–	46 mph*	–
Aurora (Microsoft) · AI	81 mi‡	98 mi NNE	6 mph	–
HAFS-B (US hurricane model)	105 mi	93 mi N	46 mph*	–
NBM · National Blend (US)	–	–	22 mph	–

Reading it: over the last two days Google DeepMind (artificial intelligence) was the most accurate on Simon's position: mean error 21 mi, and its Sat 6 pm run put the center 3 mi from where it is now. Next come Corrected consensus and NHC. HAFS-B, Aurora, HAFS-A, UKMET and AIGFS carried it 93–131 mi farther north, already over or past the bay; among them US hurricane models, which is where the hurricane-wind scenario for Vallarta came from. Locally only AIGFS got the rain right: the rest gave Vallarta more rain than fell, and almost all less wind. The sierra and the bay do not fit 15–30 mile grids.

For what is ahead, follow Google DeepMind, the NHC consensus and the ensemble means, not the ones that missed. Today they put the closest pass about 65 mi southwest of Vallarta around 11:30 pm Sunday, with a peak sustained wind of 35 mph over open water off the city (less on land); for the next 24 h the gridded models give Vallarta 0.1 in (GFS) to 0.9 in (AIGFS); GraphCast goes to 4 in, but it has been the wettest miss.

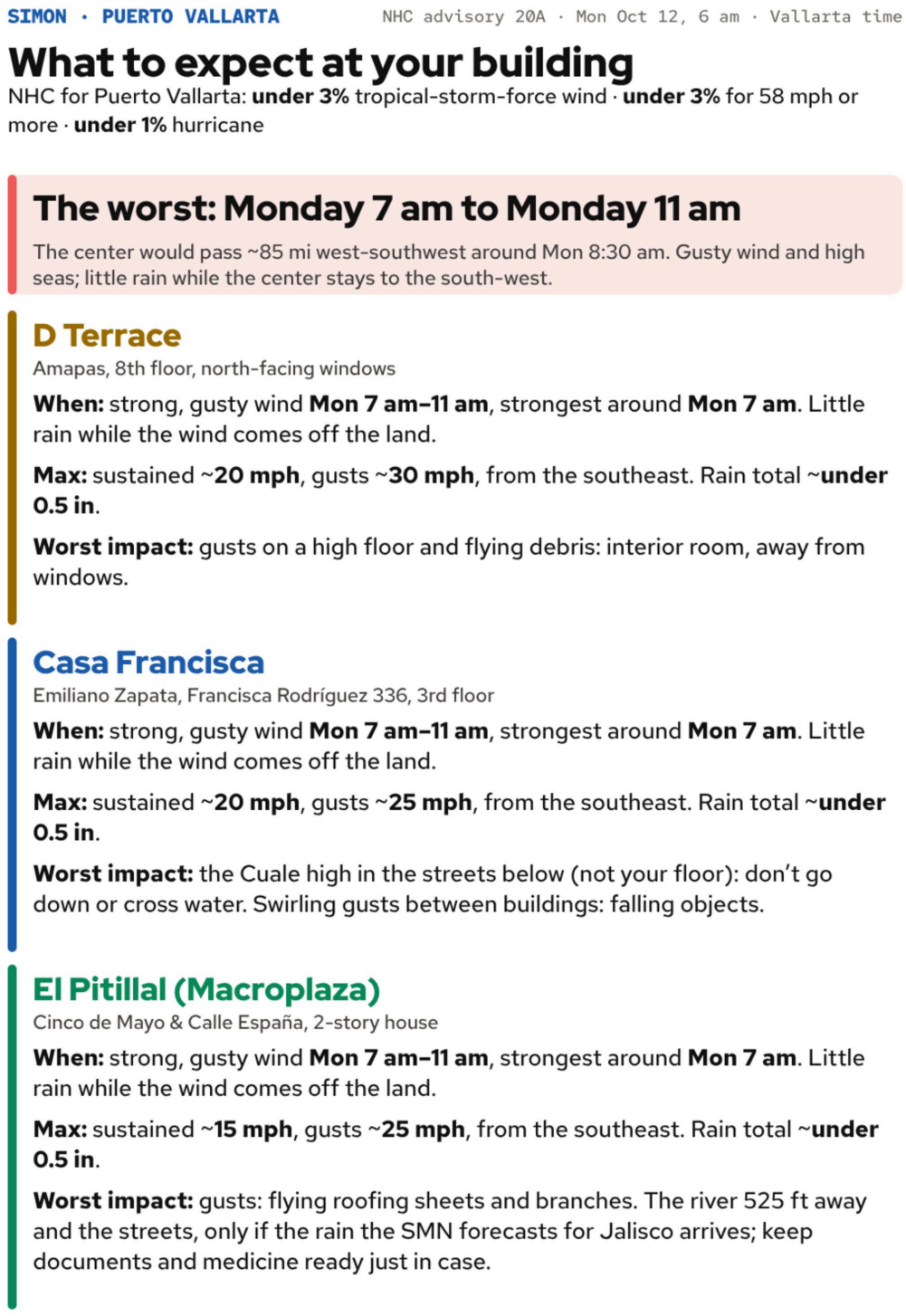
‡ Score without the 36-h lead (too few runs): not comparable with the others. † Implied wind using the model's own wind radii on its line; * using the NHC radii (track models give no point wind). Grid: wind interpolated to the city and rain at the nearest point, on 15–30 mile grids; rain accumulated from the run to 3 pm. Position: against the NHC's official track (6-hourly) and advisory 18. Sources: NHC/ATCF, ECMWF open data, NOAA (GFS, GEFS, AIGFS, NBM, HAFS), the NOAA/CIRA archive of AI models run from GFS analyses (FourCastNet, Pangu, GraphCast, Aurora), METAR MMPR.

Not an official forecast. Follow the SMN, the NHC and Protección Civil. Emergencies: 911.

A3-verify. Model verification · Advisory 18 (Sunday 3 pm).

A4 · Advisory 20 (Monday 6 am)

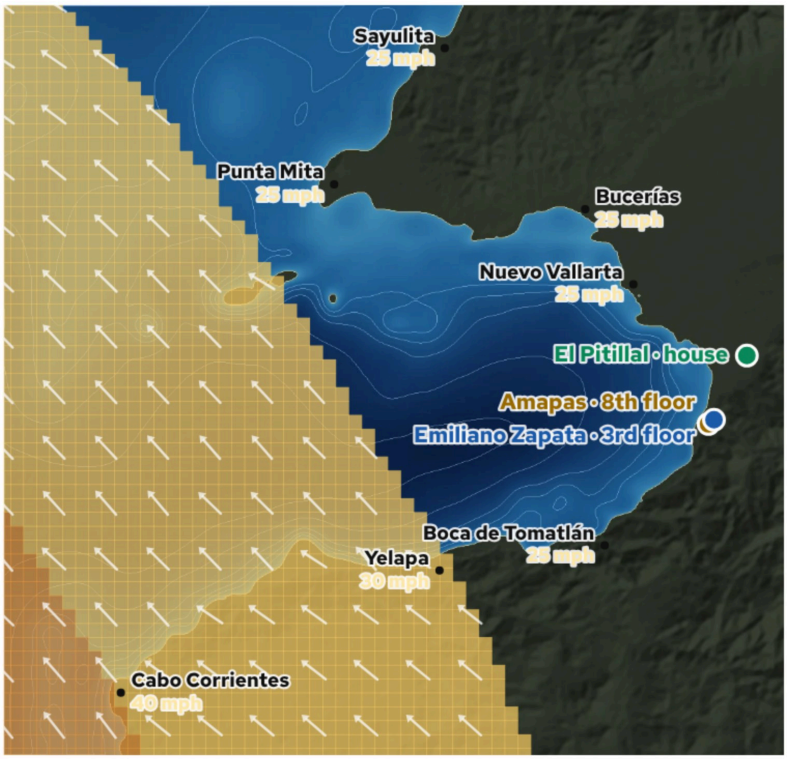
Rebuilt Monday morning with the storm stalled 120–140 km to the west-southwest; the SMN figures are from aviso 40.



A4-summary. Building summary · Advisory 20 (Monday 6 am).

Wind: how strong, from where, where it hits hardest

Colors: peak sustained wind over open water if the center follows the NHC line exactly. Arrows: where it blows toward when strongest. White line: NHC track.



29–38 mph: strong · 39–57: tropical storm · 58–73: strong TS · 74+: hurricane

Amapas, 8th floor (Cafeto & Olas Altas)

Sustained up to **20 mph**, gusts to **30 mph**, from the **southeast**.
Strong, gusty wind **Mon 7 am–11 am**. Strongest around **Mon 7 am**; the center passes around **Mon 8:30 am**.

Emiliano Zapata, 3rd floor (Francisca Rodríguez 336)

Sustained up to **20 mph**, gusts to **25 mph**, from the **southeast**.
Strong, gusty wind **Mon 7 am–11 am**. Strongest around **Mon 7 am**; the center passes around **Mon 8:30 am**.

El Pitillal, house (Cinco de Mayo & Calle España)

Sustained up to **15 mph**, gusts to **25 mph**, from the **southeast**.
Strong, gusty wind **Mon 7 am–11 am**. Strongest around **Mon 7 am**; the center passes around **Mon 8:30 am**.

Site figures: from the NHC wind odds for Puerto Vallarta (under 3% tropical storm · under 3% for 58 mph or more · under 1% hurricane) and each building's height and surroundings. They can run higher or lower. The map does not apply that correction: it shows the open-water wind if the center followed the NHC line exactly.

Not an official forecast. Follow the SMN, the NHC and Protección Civil. Emergencies: 911.

A4-wind. Wind · Advisory 20 (Monday 6 am).

SIMON · 2/6 NHC advisory 20A · Mon Oct 12, 6 am · Vallarta time

Rain: how much, and when it falls hardest

Total from now to midnight Tuesday night: today's satellite rain pattern carried along the NHC line (unscaled: the satellite sees no rain over the bay now). The official forecasts for Jalisco are far higher: see note.



Total over land to midnight Tuesday night: <4 in · 4–8 · 8–12 · 12–16 · >16 in

Amapas, 8th floor (Cafeto & Olas Altas)

Estimated total: **under 0.5 in** while the wind comes off the land.

With the center to the south-west, the wind comes over the Sierra and sinks dry into the bay: Simon's rain is over the ocean. This changes if the center comes closer or the wind turns from the south-west.

Emiliano Zapata, 3rd floor (Francisca Rodríguez 336)

Estimated total: **under 0.5 in** while the wind comes off the land.

El Pitillal, house (Cinco de Mayo & Calle España)

Estimated total: **under 0.5 in** while the wind comes off the land.

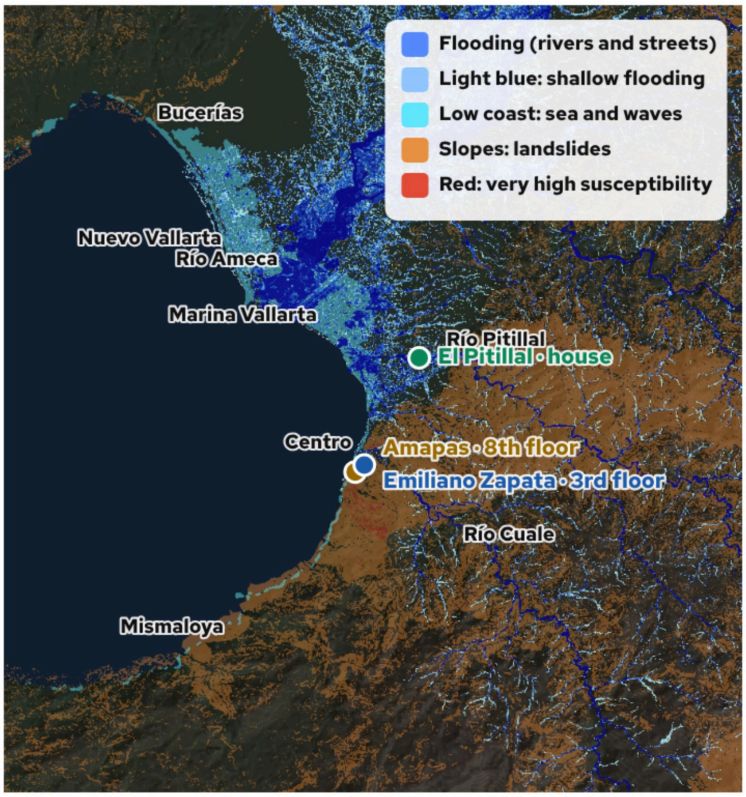
For the region: SMN (advisory 40) 2–3 in on the Jalisco coast and west, and 2–3 in in southern Nayarit (Nuevo Vallarta, Bucerías, Punta Mita, Sayulita); NHC (advisory 20A) another 2–4 in over parts of southwestern Mexico. With more rain, the rivers rise sooner.

Not an official forecast. Follow the SMN, the NHC and Protección Civil. Emergencies: 911.

A4-rain. Rain · Advisory 20 (Monday 6 am).

Water and slopes: where, and from when

Blue: flood-prone zones per CENAPRED (Ameca, Pitillal and Cuale rivers, canals). Light blue: low coast (under 13 ft). Orange: slopes highly susceptible to landslides.



Amapas, 8th floor (Cafeto & Olas Altas)

With the rain the satellite sees now, **neither the river, the streets nor the slopes reach the danger threshold** in this calculation. That changes if it rains far more than the satellite sees: the SMN still forecasts extraordinary rain for Jalisco.

Emiliano Zapata, 3rd floor (Francisca Rodríguez 336)

With the rain the satellite sees now, **neither the river, the streets nor the slopes reach the danger threshold** in this calculation. That changes if it rains far more than the satellite sees: the SMN still forecasts extraordinary rain for Jalisco.

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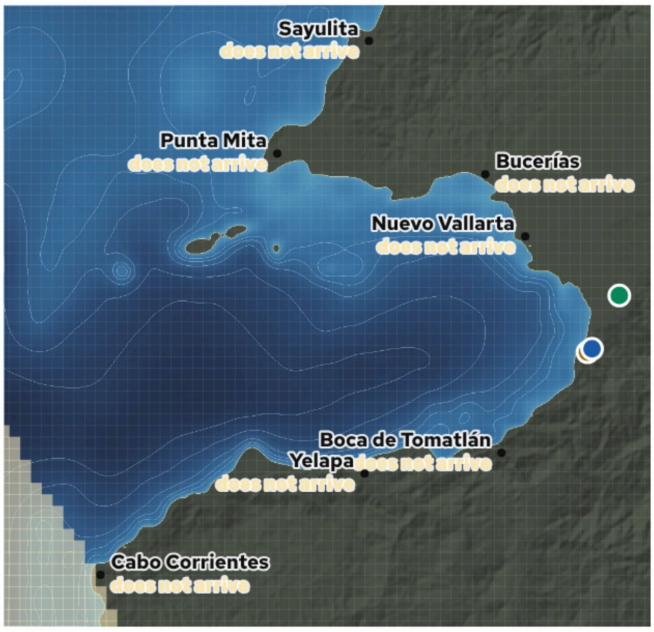
Times computed with the rain the satellite sees, carried along the NHC line (~under 0.5 in in the city). The SMN (advisory 40) still forecasts 2–3 in for Jalisco: with the center to the south-west and a wind off the land, the bay is on the dry side; if that changes, the rivers do respond. Zones: Puerto Vallarta 2024 risk atlas and CENAPRED.

Not an official forecast. Follow the SMN, the NHC and Protección Civil. Emergencies: 911.

A4-flood. Water and slopes · Advisory 20 (Monday 6 am).

When it arrives, and when it is worst

Map: when tropical-storm-force wind (39 mph) reaches each part of the bay if the center follows the NHC line. Below: hour by hour at the three sites.



39-mph wind arrives

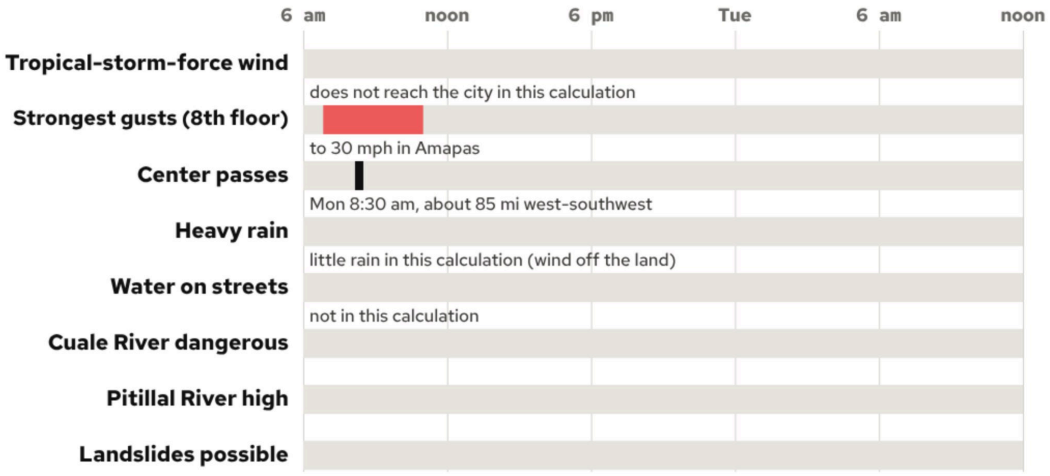
- **before Mon noon**
- **Mon noon–6 pm**
- **Mon 6 pm – Tue midnight**
- **Tue midnight–6 am**
- **Tue 6 am or later**
- **never reaches 39 mph**

Puerto Vallarta, per the NHC:

Tropical-storm-force wind: **under 3%** chance. On the NHC line it **does not reach the city**.

Hurricane-force wind: **under 1%**. The center would pass about **85 mi west-southwest**, around **Mon 8:30 am**.

What remains is the **sea**: high surf and currents around the bay. Stay out of the water.



The worst: Monday 7 am to Monday 11 am

Gusty wind and high seas; little rain while the center stays to the south-west.

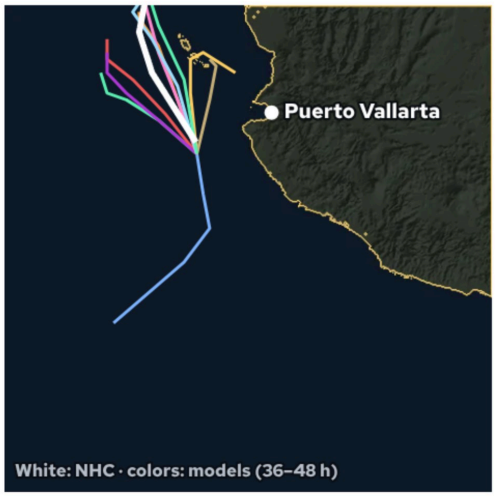
Approximate: times can shift by a few hours with each advisory. Light band: when it could start; bar: most likely. Arrow: continues past midnight Tuesday night.

Not an official forecast. Follow the SMN, the NHC and Protección Civil. Emergencies: 911.

A4-timing. Timing · Advisory 20 (Monday 6 am).

The models: which have been right, and what they say now

Score: each model's mean position error in its Simon forecasts at 12–36 h, against what happened. Bars: peak sustained wind in downtown Vallarta if the center followed that model's line.

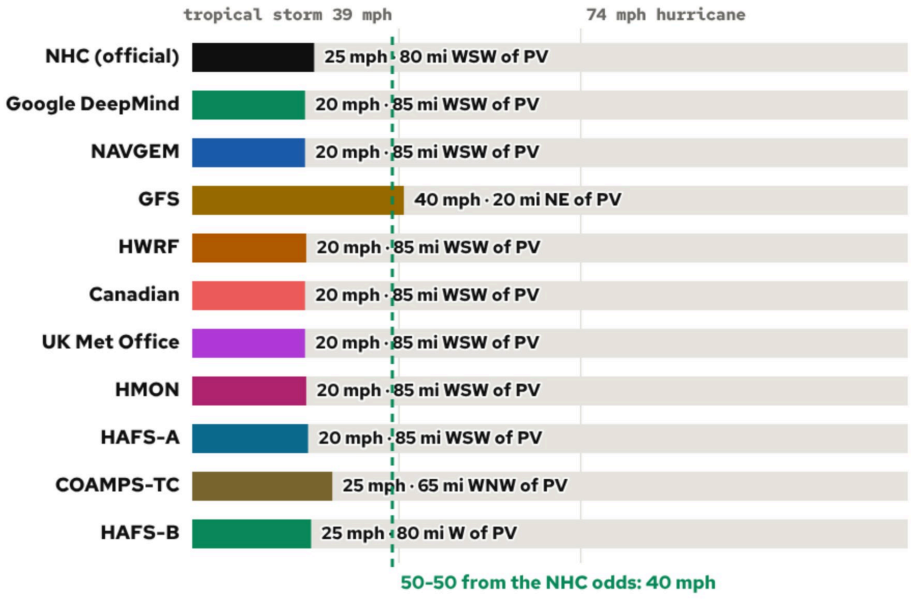


Mean position error

last 2 days · whole storm

Google DeepMind	20 mi · 30 mi
NHC (official)	30 mi · 35 mi
NAVGEM (US Navy)	40 mi‡ · 60 mi
GFS (US)	40 mi · 65 mi
HWRF	40 mi · 65 mi
Canadian	45 mi · 65 mi
UK Met Office	50 mi · 80 mi
HMON	60 mi · 60 mi
HAFS-A	75 mi · 65 mi
COAMPS-TC	80 mi · 70 mi
HAFS-B	105 mi · 65 mi

Peak sustained wind in downtown Vallarta, model by model, if the center followed its line (not the NHC odds)



Reading it: the forecasts that have tracked Simon best (Google DeepMind and the NHC) keep the center 80 mi–85 mi southwest of Vallarta. None passes it over the city. The two that bring it closest, GFS (20 mi) and COAMPS-TC (65 mi), are the ones giving 40 mph and 25 mph; over the last two days their mean error was 40 mi and 80 mi. No single model decides: the NHC gives Vallarta **under 1%** for hurricane-force wind and **under 3%** for tropical-storm force.

Models from the 12Z 12 Oct cycle (NHC/ATCF). Error: against the observed official track; "last 2 days" averages only 1 to 7 forecasts per model and lead time, so the order can change. Vallarta wind: the NHC wind structure on each model's line, without the odds-based correction.

Not an official forecast. Follow the SMN, the NHC and Protección Civil. Emergencies: 911.

A4-models. Models · Advisory 20 (Monday 6 am).

Appendix B. Methods and sources

Position verification

Forecasts from every cycle between 3 pm Friday and noon Sunday (09/21Z to 11/18Z), verified up to 3 pm Sunday (11/21Z), were interpolated to the NHC best-track times (6-hourly) and to the advisory-18 position (11/21Z, 19.6N 105.8W). Errors are great-circle distances. The score is the mean error at 12, 24 and 36 h lead. Track-type models come from the public ATCF a-deck (late versions: AVNO, CMC, UKX, NVGM, HWRF, HMON, HFSA, HFSB, CTCX, GDMN, CEMN, AEMN, HCCA, TVCN, CLP5, OFCL); gridded models were tracked as the minimum sea-level pressure inside 15–24N, 110–101W, with points on the box edge discarded. ECMWF IFS, AIFS and the two ECMWF ensembles come from the open-data tropical-cyclone track files (BUFR) and GRIB fields; GFS, GEFS mean, AIGFS, NBM and HAFS-A from the NOAA open-data buckets; FourCastNet v2, Pangu-Weather, GraphCast and Aurora from the NOAA/CIRA archive of AI models run from GFS analyses (no 6 pm Friday run exists there, which is why their scores lack a 36-h lead).

Intensity

Track models: the a-deck maximum wind. Gridded models: the strongest 10-m wind within 2° of the model's own low. Whole-storm intensity errors are the mean absolute error at 12–36 h over all cycles of the storm.

Vallarta wind and rain

Observations: METAR MMPR (airport) for the 48 h to 3 pm Sunday; there are no reports between 6 pm Saturday and 1 pm Sunday, and no official gauge in the city. Gridded models: bilinear interpolation of the 10-m wind to 20.61N 105.23W; rain accumulated from the run start at the nearest grid point, with the maximum of the four surrounding points also kept. Track models: a parametric wind field with the NHC wind radii of the same cycle, or the model's own radii where the deck carries them, evaluated at the city over open water, before any land or building exposure factor.

The resident graphics

City wind: the parametric wind field on the NHC line evaluated at the city, capped by the wind value at which the NHC wind-speed probabilities give 50%, then scaled by exposure factors for each building (0.89 for the exposed high floor with an east wind, rising to 1.08 when the wind comes off the bay; 0.77 for the sheltered mid floor; 0.70 for the street-level house). Bay-wide fields: a parametric wind model on the NHC line with the official radii; rain from the GOES-18 rainfall-rate product carried in a storm-relative frame along the NHC line, calibrated only downward, never up. Flooding and slope layers: Puerto Vallarta risk atlas (2024), CENAPRED, terrain tiles. All times are Vallarta time (UTC−6).

Forecast evolution and outcome

Figures 2–3 use the NHC forecast/advisory products for advisories 1–20: each advisory's forecast points were interpolated hourly and the closest approach to 20.61N 105.23W taken as the minimum distance; the intensity is interpolated at that time. The real closest approach uses the NHC best track to Monday 6 am interpolated the same way. Figure 5 uses the building summaries' own numbers (bay.json for each version) and the airport reports (METAR MMPR) from Sunday 3 pm to Monday 8 am. Print versions of the resident graphics (Appendix A) are the same renderer with page colours remapped for a white background; map areas keep their original colours. From the Monday build onward, satellite centre fixes more than 60 km from the NHC analysed or interpolated position are replaced by the NHC position before the fields are computed (section 4).

Sources

NHC forecast/advisory, discussion, wind-speed probabilities and forecast advisory for EP202026 (advisories 1–20A); NHC ATCF a-deck and b-deck; SMN-CONAGUA avisos de ciclón tropical (through aviso 40); ECMWF open data (IFS 0.25°, AIFS single, IFS ENS, AIFS ENS); NOAA NOMADS/AWS open data (GFS 0.25°, GEFS, AIGFS, NBM v4 oceanic domain, HAFS-A); NOAA/CIRA ML weather prediction archive (Radford et al. 2025, BAMS); NOAA GOES-18 RRQPE; aviationweather.gov METAR; OpenStreetMap; AWS Terrain Tiles; Atlas de Riesgos de Puerto Vallarta 2024; CENAPRED. The uploaded analysis is "Comprehensive Verification and Theoretical Analysis of Dynamical and AI-Driven Tropical Cyclone Forecasting: The Case of Hurricane Simon and the Epistemology of Risk Communication" (13 pages, undated).

Sources for section 10

Storm histories: NHC tropical cyclone reports for Kenna (EP142002), Patricia (EP202015), Roslyn (EP192022), Lidia (EP152023), Otis (EP182023), John (EP102024), Erick (EP052025) and Priscilla (EP162025), nhc.noaa.gov/data/tcr; NHC advisories and discussions for Simon (EP202026). Model performance: NHC Forecast Verification Report 2025 (nhc.noaa.gov/verification); NHC model summary (nhc.noaa.gov/modelsummary.shtml); NOAA/NWS, "AI hurricane forecasting" (weather.gov/news/261102-AI-Hurricane-Forecasting); DeMaria et al. 2024, arXiv:2409.06735; Allen et al. 2026, "AIFS-TC", arXiv:2608.09959; Subich et al. 2025, arXiv:2501.19374; Google DeepMind, Weather Lab (June 2025) and WeatherNext Cyclones (Nature, August 2026); ECMWF, AIFS operational announcements (February and July 2025); NOAA, AI global models announcement (December 2025); NVIDIA Earth-2 (January 2026); Local10/WPLG, 10 September 2026 (season-to-date verification reported by James Franklin). Climate: IPCC AR6 WG1 SPM, Chapter 11 and Chapter 12 (2021); Knutson et al. 2020, Bull. Amer. Meteor. Soc.; Balaguru et al. 2024, Earth's Future, doi:10.1029/2023EF004230; Radfar et al. 2026, Science Advances, doi:10.1126/sciadv.adu1733; Murakami et al. 2017, J. Climate; Calderón-Vega et al. 2026, J. Mar. Sci. Eng. 14(8):706. Observing network: Conagua/SMN inventory presented to the CEC, September 2019; Auditoría Superior de la Federación, performance audit 2024-5-16B00-07-0103-2025 of programme E006; El Universal, November 2020 (radar network); NCAR/UCAR NAME archive (SMN radar sites); Universidad de Guadalajara (IAM radar); National Hurricane Operations Plan 2022 (NHOP, section 5.4.1.2.2); 53rd Weather Reconnaissance Squadron 2025 season summary (DVIDS, December 2025); Secihti joint communiqué on Apixqui, 9 June 2026, and Crónica, 24 May 2026; presidential briefing of 29 June 2026 as reported by Zeta Tijuana. Institutions: Manual SIAT-CT 2019 (gob.mx); CENAPRED, "Qué hacemos"; Ley de Protección Civil del Estado de Jalisco; WMO RA-IV Workshop 2023 introduction; WMO–SMN meeting of 19 March 2020; WMO Geneva Declaration 2019; WMO and UNDP on SOFF and Early Warnings for All (2022–2026). Finance: World Bank press release, 15 May 2024 (Mexico Pacific cat bond); Artemis.bm, 10 June 2026 (parametric renewal), 12 October 2026 (Simon and the cat bond trigger) and on FONDEN's closure; AMIS insured-loss estimate for Otis (El Universal, January 2024); World Bank project P126487 (MoMet); IDB, Alliance for Hydromet Development; NOAA NESDIS Commercial Data Program award, 13 August 2026; Met Office–Microsoft agreement, April 2021; Quintana Roo coastal zone trust and reef insurance (NBC, earth.org). Local: Asociación de Hoteles de Puerto Vallarta and Coparmex Puerto Vallarta statements during Simon (Mexican Times, Tribuna de la Bahía, Global Media, 9–11 October 2026); Gaceta UdeG on the 2016 Puerto Vallarta risk atlas. Every figure quoted in section 10 was checked against the source listed; where a source could not be reached directly it is marked in the text as reported.

Shared graphics page (Spanish and English): https://claude.ai/artifact/Xw3VMEvngPTbxLXkRDNvYd · Street map of the bay: https://claude.ai/artifact/8nN9cLcj6DWw4FGycv9xyj · Code and data: branch claude/hourly-forecast-animation-m4v1bj of the vallarta-weather repository. Not an official forecast. Follow the SMN, the NHC and Protección Civil. Emergencies: 911.